



**KTH Industrial Engineering
and Management**

Electrified Africa - Associated investments and costs

Author:

Ioannis Pappis

pappis@kth.se

MSc Sustainable Energy Engineering

Supervisor:

Mark Howells

Director and Professor: Division of Energy Systems
Analysis

Royal Institute of Technology, Sweden (KTH-dESA)

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KTH School of Industrial Engineering and Management
Division of Energy Systems Analysis
SE-100 44 Stockholm



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Ioannis Pappis

Approved	Examiner	Supervisor
11 October 2016	Mark Howells	Mark Howells
	Commissioner	Contact person pappis@kth.se

Abstract

Africa is a resource rich continent in both fossil fuels and renewable energy sources. Nevertheless, its significantly underdeveloped electricity infrastructure and un-deployed power generation potential lead to more than 620 million people to lack electricity access and the rest to face shortages and high prices. Africa's rapidly increasing electricity demand, from 621 TWh in 2012 to about 1870 TWh in 2040, as well as rapid population growth, make urgent the necessity for national and regional investments which will lead on reliable and affordable energy widely available and achieve social and economic development. This thesis examines the national associated costs and investments which are needed to electrify the current un-electrified population of the continent by 2030 and provide them to have an electricity consumption of 696kWh/household (TIER 3) that year and 2195kWh/household (TIER 5) in 2050, as well as the continent to cover its electricity demands for the period 2010-2050. To conduct this study, The Electricity Model Base for Africa (TEMBA) [1] developed in the Open Source energy Modelling SYstem (OSeMOSYS) [2] is used, with the initial modeling platform to remain the same but several parameters to be updated to the latest up-to date source and the objective of the TEMBA study to differentiate. The parameters which have been updated in this study for each country include: electricity demands (2010-2050), existing and planned power plant capacities, technology costs, fuel availability and prices, national fuel reserves, local transmission and distribution losses, and renewable resources potential. Moreover, emission factors have been added into the model to estimate greenhouse gas emissions. Different trade scenarios have been implemented to investigate the power generation potential and the financial requirements which are needed by the continent to cover its electricity demands. An expanded electricity trading scheme in Africa, as is being indicated in the Enhanced scenario, exploits its country's energy resource potential and lead to a decrease of the investment and fuel costs as well as of the carbon dioxide emissions. The open-source nature of the model allows the data and the model under which this study is being conducted to be publicly available for future research.

Acknowledgments

“You can make many speeches, but the real thing is when you dig a hole, plant a tree, give it water, and make it survive. That’s what makes the difference.” Wangari Maathai (1940-2011), the first African woman to receive the Nobel Peace Prize. This is the epitome for me, and with the outcome of this thesis, I hope I will make a difference on more levels than just the theoretical.

First, I would like to thank the supervisor of my thesis Prof. Mark Howells, director of Energy Systems Analysis (dESA) at the Royal Institute of Technology in Sweden, for his motivation, wide knowledge in the energy field and with extensive experience in the energy sector of Africa, encouragement, and valuable advices all this time.

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1. Introduction

Africa is endowed with the widest possible range of energy resources all over the world, however its electricity generation installed capacity is almost 165 GW and only 20% comes from renewable technologies [3]. The power generation installed capacity of the world's second-largest and second-most populous continent is almost the same as the one the United States of America installs in five years with a population of approximately one quarter of Africa's or China installs in approximately two-three years [4]. The significantly underdeveloped electricity infrastructure and un-deployed power generation potential of the continent leads to more than 620 million people to lack electricity access, around 50% of Africa's total population, and the rest to face shortages and high prices [5]. The low electrification rates, 68% in urban and 26% in rural areas, nearly 60% of Africa's population to live in rural areas [5], associated with Africa's significant projected population growth and energy demand, 1.18 billion people and 589 TWh [6] [7] in 2013, and approximately 2.01 billion people [8] and 1869 TWh in 2040 [9], make urgent the need for investments in the energy sector to provide reliable and affordable energy widely available and environmentally sustainable. Towards that direction and fully aware of continent's energy sector challenges, forty-seven African Member States signed the Paris Agreement under the United Nations Framework Convention on Climate Change on 22 April 2016 [10] [11]. The outcomes of this agreement, source funding, technology transfer, climate finance and capacity development is expected to help Africa to develop its renewable energy resources and mitigate climate change. Also, two out of six pillars, science, technology and innovation, and environmental sustainability natural resources management; and disaster risk management, stated in the Common African Position (CAP) on the Post-2015 Development Agenda by the African leaders, indicate the development priorities in the continent [12]. The UN Secretary-General Ban Ki-moon's launch of Sustainability Energy for all in September 2011, the United States of America President Obama's launch of Power Africa, and the UK's Energy Africa campaign in 2015, are some additional significant steps which have been done to provide help in Africa to face its challenges.

The objective of this thesis is to investigate the power generation potential and the financial requirements which are needed to electrify the currently un-electrified population of Africa, considered future newly electrified people and their energy intensity, as well as to meet the electricity demands during the period 2010-2050. The electricity supply system of forty-seven African countries modelled individually (TEMBA) in an open source long-term cost-optimization tool (OSeMOSYS) to examine the less costly generation mix of each country has been updated [1]. Moreover, two trade scenarios have been implemented to investigate the effects of the international grid interconnections in each country's power generation infrastructure. Investments towards that direction, extensive grid interconnections among the countries, could exploit each country's potential energy resources and lead to less fuel and investment costs, lower electricity prices, and less carbon dioxide emissions.

2. Overview of Methodology

2.1 Methodology

The Electricity Model Base for Africa (TEMBA) developed in the Open Source energy MOdelling SYstem (OSeMOSYS) [1]. TEMBA includes the electricity supply systems of forty-seven African countries, modeled individually and linked via trade links, to develop least cost system configurations the period 2010-2050. The "least cost" is defined for the entire system, and for the entire modeling period - not for each country individually.

In this thesis, the initial modeling platform remained the same but several parameters are updated to the latest sources and the objective of the TEMBA study differentiated. The objective of this thesis is to electrify Africa, except from covering each country's electricity demands the period 2010-2050 as are included in the TEMBA study. The parameters which have been updated in this study for each country are the following: electricity demands (2010-2050), existing and planned power plant capacities, technology costs, fuel availability and prices, national fuel reserves, local transmission and distribution losses, and renewable resources potential (hydro, solar, geothermal). Moreover, emission factors have been added into the model to estimate greenhouse gas emissions. The countries which are included in the analysis, forty-seven in total, have been divided in five power pools, and are presented in Figure 1. South Sudan and Sudan have been included in the model as one country, and Sao Tome and Principe, Madagascar, Mauritius, Seychelles, Cape Verde have been excluded from the study since they are island states that do not affect trade potential within the continent.

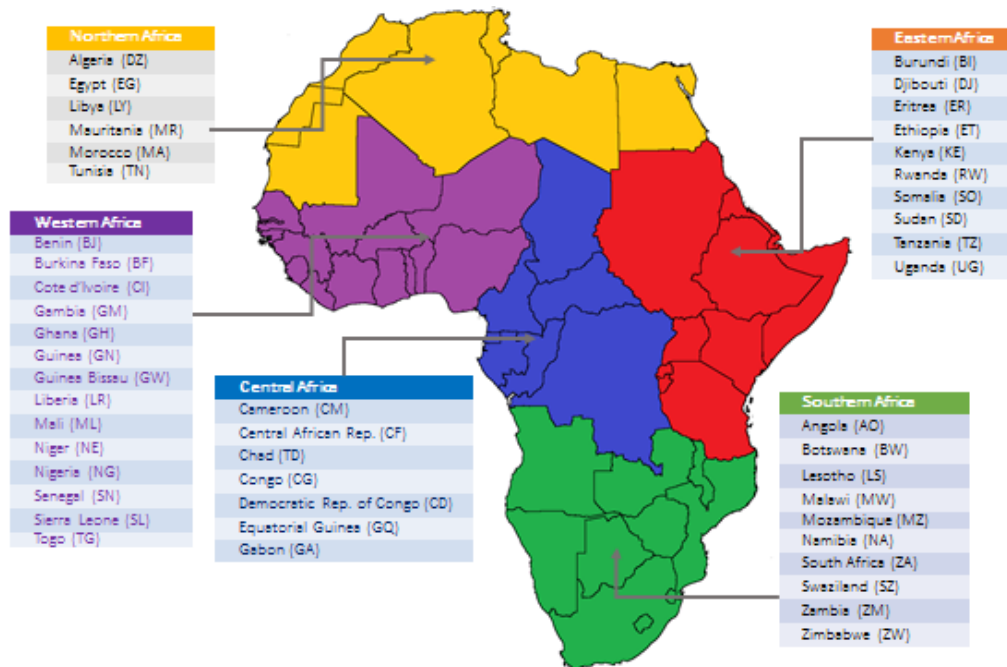


Figure 1 Countries included in the model by sub-region.

2.2 The Open Source Energy Modelling System (OSeMOSYS)

OSeMOSYS is a dynamic, bottom-up, full-fledged systems optimization model for long-run energy planning [2]. Unlike long established energy systems (partial equilibrium) models, such as MARKAL/TIMES (ETSAP (Energy Technology Systems Analysis Program), 2010), MESSAGE (IAEA (International Atomic Energy Agency), 2010), PRIMES (NTUA (National Technical University of Athens), 2010), EFOM (Van der Voort, 1982) and POLES (Enerdata, 2010), OSeMOSYS learning curve and time commitment to build and operate is significantly low. It builds on an open source programming language (GNU MathProg) and uses the GNU Linear Programming Kit (GLPK) solver, therefore, no upfront financial expenditures are needed. OSeMOSYS objective function is to define the minimum operational cost of the system, by investigating the production

mix of technologies and fuels which required to meet the given energy demand, while satisfying all the constraints which have been setted for the modelling period. Its block structure (Figure 2) and flexibility, as well as its open source nature, allow the energy analyst to define his/her model complexity and to elaborate a wide range of applications.

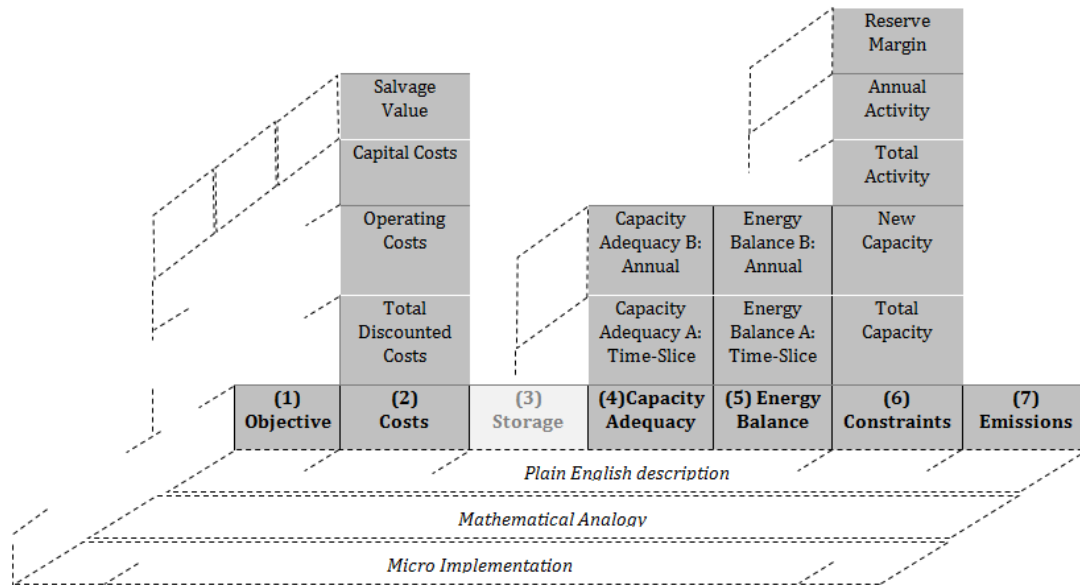


Figure 2 OSeMOSYS 'blocks' and levels of abstraction [2].

2.3 The Reference Energy System (RES)

The schematic representation of the electricity supply system of each country included in the modelling framework is represented in Figure 3. In the RES, the technologies are being represented by boxes and the fuels by lines. Techno-economic (power plant capacities, capacity factors, efficiencies, lifetime, costs) and environmental (CO_2 emissions) parameters are assigned to the technologies. Each final demand, has been divided into three sectors, and is being supplied by different energy chains based on the least cost option for the whole continent computed by the model the timeframe 2010-2050. The extraction or imports of fuel resources accordingly to each country's fuel availability define the "primary" level. The fuels which are supplied to the power plants in the "secondary" level, transformed to generate electricity which will be transmitted through the national transmission network to the "tertiary" level, in order to be distributed and meet each sector's final demand. De-centralized technologies, unlike centralized ones, are not connected to the national grid but generate and supply electricity directly to end-users. Also, in the model electricity imports/exports from one country to another have been taken into account as are presented in the Figure 3 below.

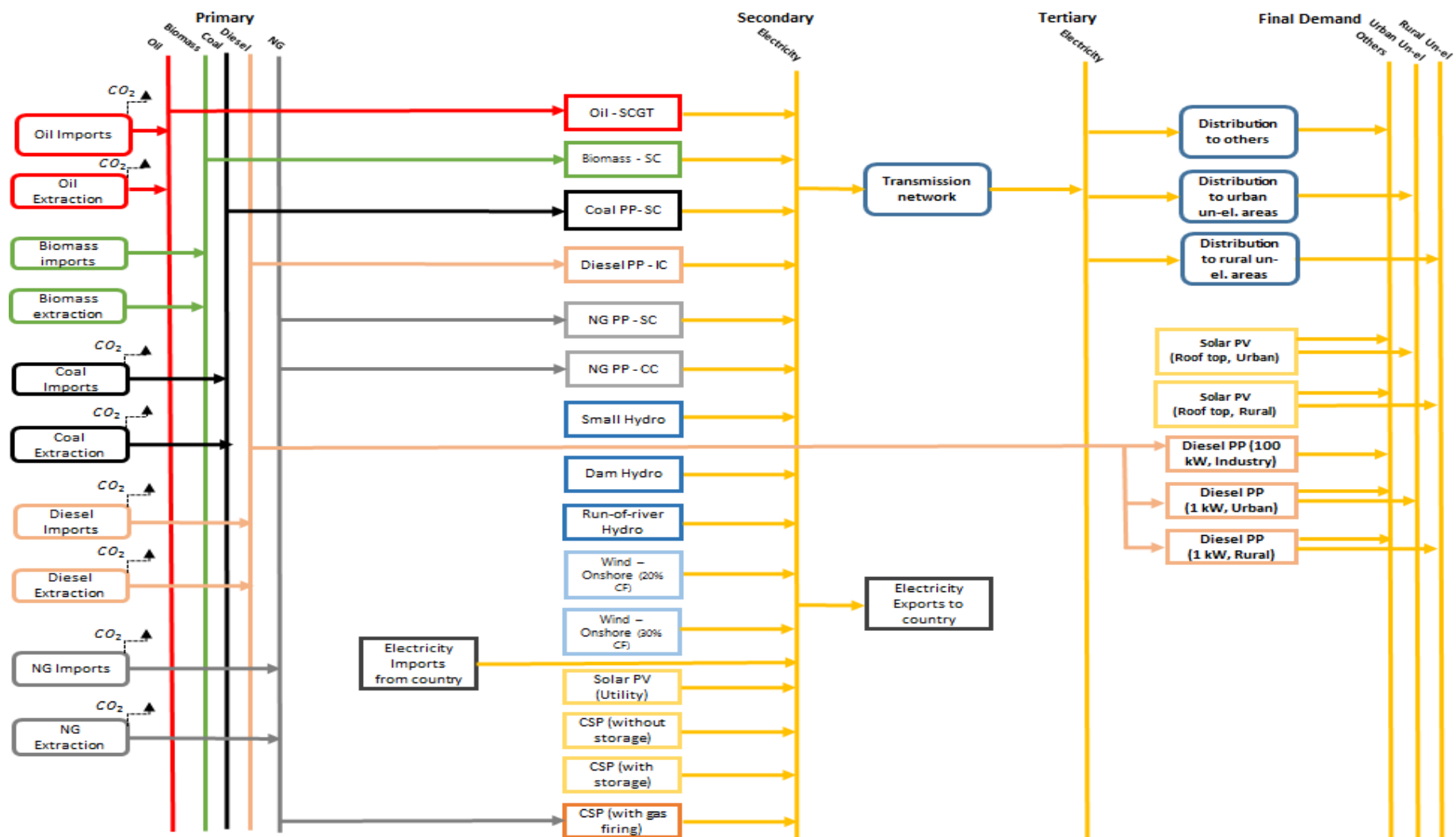


Figure 3 Country power sector model structure.

3. Scenario Description

Scenario analysis is essential for energy and investment planning, financing needs, informing and formulating energy policies, strategies, debates, decisions, as well as for academic purposes. In this thesis two trade scenarios, Reference and Enhanced, are assessed to investigate the effects of the international grid interconnections in each country's power generation infrastructure the period 2010-2050. The expansion of the currently limited grid interconnections among the African countries could exploit each country's energy resources and lead to potential benefits for the continent. The grid interconnection projects which are included in the model and assigned in each country in each scenario are provided in Table 17 of Appendix C.

- ♦ The **Reference scenario** includes the cross-border electricity interconnections among the African countries which exist and are committed (Table 17, Appendix C). In this scenario, no future potential trades are considered in the model. The future electricity generation mix as well as the investments which are needed by the continent to cover its electricity demands are examined based on the existing energy infrastructure.
- ♦ The **Enhanced scenario**, additional to the existing and committed trades, includes the planned uncommitted grid interconnections. Potential cross-border electricity interconnections projects among neighboring countries are considered in the model after 2025 to investigate the least cost electricity generation mix of the system.

4. Model Assumptions

4.1 Overall Assumptions

Overall assumptions across all scenarios are as follows:

- ♦ The real discount rate is 5%.
- ♦ The monetary unit is 2010 US\$.
- ♦ The timeframe of the model spans from 2010 to 2050, with yearly basis simulations.
- ♦ In order to capture the key features of electricity demand load pattern, each year has been splitted in two seasons, and each season in two blocks of equal demand, namely "day" and "night".
- ♦ Transmission and distribution losses are defined on a national level. Different efficiency improvements are assumed to be achieved during the model horizon based on each country's current status (Table 16, Appendix C).

4.2 Electricity Demand

The main sources used for each country's historic electricity demand (until 2013) are the Energy Statistics and Balances of International Energy Agency [6] and the 2013 Electricity Profiles published by United Nations Statistics Division [7]. National indicators, such as urban and rural population [8], electrification rates [5], and energy intensities are used to derive the final electricity demands. The electricity demand of Africa was 540.4 TWh in 2010 and based on the assumptions of this study will be 3416 TWh in 2050. The values of each country's electricity demand, after transmission and distribution losses, can be found in Table 9 of Appendix A. Also, the number of the un-electrified people and each country's electricity access in 2016 considered in the model are presented in the Figure 4 below.

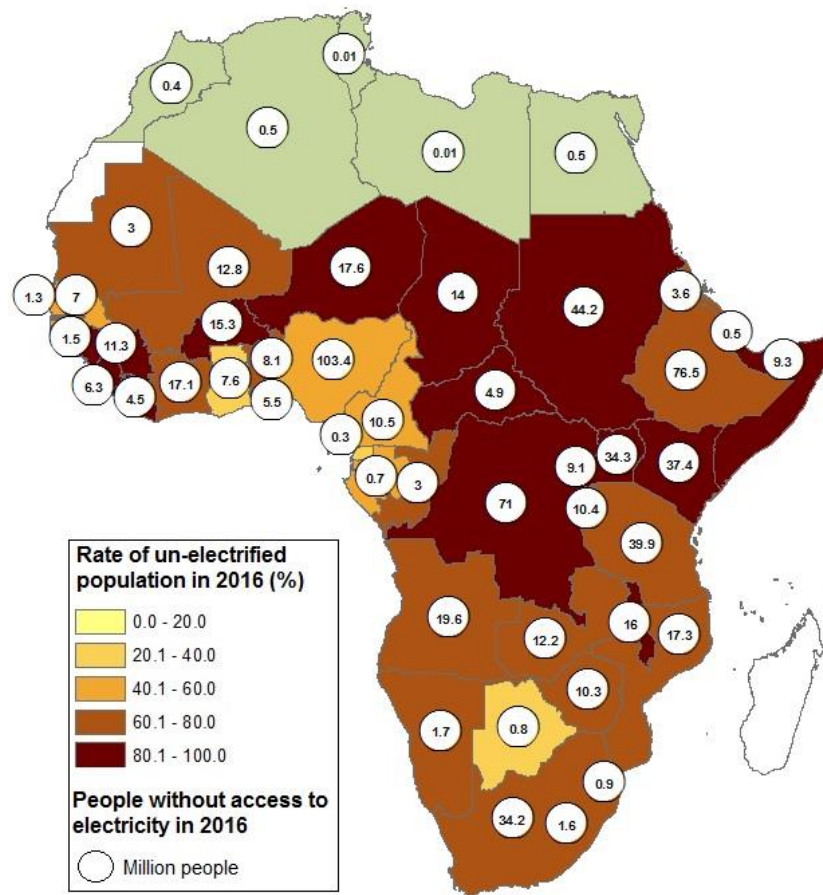


Figure 4 Number and share of population without access to electricity by country in 2016.

The methodology which has been applied to calculate the electricity demand projections of each country remains the same in all cases. The demand has been divided into “urban un-electrified”, “rural un-electrified” and “others” categories as follows:

- ♦ **Urban un-electrified**, the urban population who is currently un-electrified and gains electricity access gradually through the period 2017-2030;
- ♦ **Rural un-electrified**, the rural population who is currently un-electrified and gains electricity access gradually through the period 2017-2030;
- ♦ **Others**, refers to all the currently electrified portions of sectors in the economy. This includes industry, transport, commercial and public services, agriculture, and residential.

The demand projection of “others” sector is based on the projected relative increase among the years derived from the African Energy Outlook 2014 [5] for each power pool (Table 12, Appendix A). The timeframe of the model spans after 2040, so the growth rate of period 2030-2040 is applied for the years 2041-2050.

The “urban and rural un-electrified” demands represent the urban and rural population who is currently un-electrified based on the electrification rates of 2012. It has been assumed that the electrification rate of each country will be 100% in 2030, and the energy intensity for both urban and rural population will be 696 KWh/household (TIER 3) in 2030 and 2195 KWh/household

(TIER 5) in 2050 [13]. Each household has been assumed to correspond to five persons. The electrified population considered in the model for the period 2016-2050 as well as the electrification rates of Africa in 2012 can be found in Table 10 and Table 11 of Appendix A correspondingly.

The total demand of the continent is increasing approximately six times 3416TWh in 2050, compared to 540.4TWh in 2010 (Table 1). All the population of Africa is electrified in 2030 and have an electricity consumption of TIER 3. That year (2030) the total demand of the continent is approximately two times higher than the 2010. The total demands of Central and Western African power pool in 2030 are approximately three times higher than in 2010. This is mainly due to Democratic Rep. of Congo and Nigeria's low electrification rates and high population. The electricity demands of these two countries constitute around 50% of their power pool's total demands in 2015. The Eastern African power pool represents the highest increase, approximately by six times, of its total electricity demand in 2030. This is mainly due to Ethiopia, Kenya and Sudan where these countries constitute around 60% of the power pool's total demand.

Table 1 Demand per power pool in TWh.

TWh	2010	2015	2020	2030	2040	2050
CAPP	13.6	18.9	28	68.9	138.5	244.8
EAPP	22.2	38.1	62.3	180.7	378.3	698.0
NAPP	222.3	278.8	340.4	450	594.8	739.3
SAPP	242.6	256.4	295.6	400.8	552.4	760.5
WAPP	39.8	58.0	89.2	230.7	497.2	973.5
Africa	540.4	649.8	815.4	1331.0	2143.1	3416.0

The northern African countries have almost 100% electricity access, except from Mauritania which has 21%, consequently their urban and rural un-electrified demands are relatively low. Countries in West and East African power pool account for most of the un-electrified demand since they have low electrification rates and high population. Figure 5 shows the evolution of the electricity demand by sector in Africa the period 2010-2050. The total electricity demand in 2050 will be approximately 6 times higher than the 2013 values. The share of the population that gain electricity access in 2020 is approximately 2% of the total demand that year, and increases to 23% in 2050.

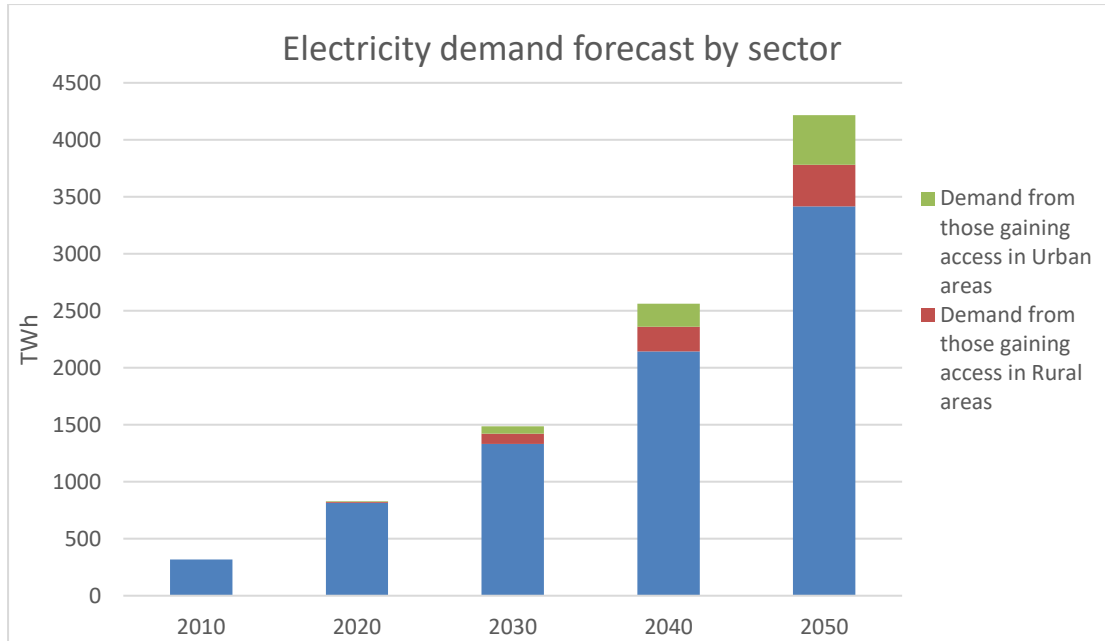


Figure 5 Projected electricity demand in Africa and the share of those that gain access.

4.3 Power generation technologies

In an energy system, in order for the final demand to be covered, generic power generation technologies are modeled to provide electricity through the grid. These technologies are being categorized mainly in renewable and non-renewable, and in centralized (upstream of transmission) and decentralized (upstream of distribution or site specific). Also, certain techno-economic parameters are defined for each technology [9] [14] [15] [16].

Generic Renewable Power Generation Technologies

The following renewable technologies are included into the model in addition to geothermal and nuclear. Nuclear technology is avoided to be considered as a viable energy option into the model, is available only in South Africa (there is already 1.8 GW of installed capacity) and Egypt, while new additions of this technology can be done only from 2026 onwards. Only the distributed or roof-top solar PV technology is decentralized, while all the other technologies are connected upstream of transmission.

- **Small hydro** refers to the power plants which their capacity is less than 20 MW.
- **Medium hydro** represents the power plants with capacity which range between 20 – 100 MW.
- **Large hydro** represents the run of river and reservoir options for power plants with capacity more than 100MW.
- **On-shore wind** is represented by two technologies based on their capacity factors, 25% and 30%.
- **Biomass** include the co-generation technologies.
- **Utility PV** represent the PV farms which are managed by the utility. They are modeled to produce electricity only during the day.
- **Distributed or roof-top Solar PV** are modeled to produce electricity only during the day. They can supply urban-rural residential, commercial, and small industries demands.

- **Solar CSP no storage** represents medium to large scale concentrated solar power plants. They were modeled to produce electricity only during the day
- **Solar CSP with storage** are medium to large scale concentrated solar power plants with thermal storage. They can produce electricity throughout the day.

The learning rates of the renewable technologies throughout the model period are anticipated to be improved. It has been assumed that from 2035 onwards the costs remain constant. The projection of the overnight investment costs of the renewable technologies are presented in Figure 6. Also, the techno-economic parameters of the renewables and fossil fuel power generation technologies used in the model are presented in Table 2.

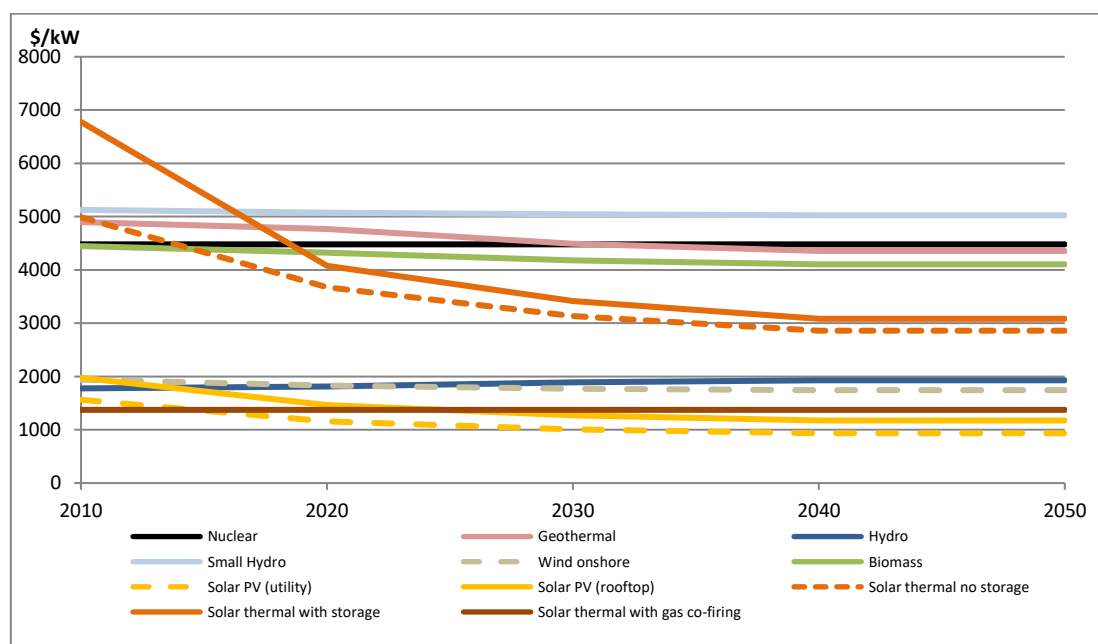


Figure 6 Overnight investment costs for renewable technologies.

Table 2 Techno-economic parameters for power generating technologies [9] [14] [15] [16].

Technologies	Overnight capital cost (USD/kW)			Fixed cost (\$/kW)		Variable cost (\$/MWh)		Life (Yrs)	Efficiency	Load factor	Construction time (Yrs)		
	2010	2020	2035	2010	2050	2010	2050				2010	2020	2035
Diesel centralized	708	708	708	8	8	2	2	25	35%	80%	2	2	2
Diesel 100kW system	659	659	659	-	-	55	55	20	35%	80%	1	1	1
Diesel/Gasoline 1kW system	692	692	692	-	-	33	33	10	21%	72%	1	1	1
HFO	1350	1350	1350	-	-	15	15	25	35%	80%	2	2	2
OCGT	708	708	708	8	8	2	2	25	30%	85%	2	2	2
CCGT	1021	1021	1021	1	1	2	2	30	48%	85%	3	3	3
Coal	2080	2080	2080	27	27			35	37%	85%	4	4	4
Hydro small scale	5127	5075	5025	41	40	7	7	50	N/A	Varies	4	4	4
Hydro medium scale	1777	1815	1929	26	26	5	5	50	N/A	Varies	4	4	4
Hydro large scale	1777	1815	1929	22	24	4	4	50	N/A	Varies	4	4	4
Biomass	4447	4325	4105	59	54	3	3	30	38%	50%	3	3	3
Nuclear	4480	4480	4480	112	112	2	2	60	33%	85%	7	7	7
Geothermal	4898	4769	4356	49	44	-	-	25	100%	85%	4	4	4
Wind onshore	1940	1829	1743	37	33	-	-	25	N/A	Varies	2.5	2	1.5
Solar PV (utility)	1562	1158	935	25	22	-	-	25	N/A	25%	1.5	1.5	1.5
Solar PV (rooftop)	1977	1461	1175	33	29	-	-	25	N/A	20%	1	1	1
Solar CSP no storage	4988	3677	2860	200	114	-	-	25	N/A	35%	3	2.8	2.5
Solar CSP with storage	6777	4074	3083	61	42	4	2	25	N/A	63%	3	2.8	2.5
Solar CSP with gas co- firing	1374	1374	1374	-	-	16.4	16.4	25	53%	85%	3	2.8	2.5

4.4 Technology emission factors

Fossil fuel technologies based on the fuel which are using to generate electricity emit different levels of greenhouse gases which are harmful to the environment. In the model, different carbon dioxide emission factors have been linked to each fuel, on the output of import and extraction technologies in each country, and are presented in Table 3 [17]. These linkages help to identify the total amount of carbon dioxide emissions being produced by each country.

Table 3 Carbon dioxide emission factors per fuel [17].

Fuel	Mton/PJ
Diesel	0.0693
HFO	0.0747
Gas	0.0503
Coal	0.0893

4.5 Fuel Availability and Prices

The categorization of the fuel prices has been done based on the location of each country (inland, coastal), as well as their availability on fuels (domestic reserves, imports). The prices of the imported fuels for the inland countries, compare to their domestic ones, were calculated based on their distance from the closest coastal capital. The distance coastline factors of each country can be found in Table 15 of Appendix B. Each country's proven reserves in coal (bituminous, lignite), crude oil and natural gas, which were added in the model can be found in Table 13 of Appendix B [4].

Coal was/is distinguished based on its different energy content, in bituminous 27-30 GJ/ton, and lignite 10-19 GJ/ton. Therefore, different prices are assumed for each country's identified domestic reserves. Morocco and Nigeria which have bituminous and lignite reserves, a weighted average price was calculated for the coal domestic price.

Two types of crude oil products (heavy fuel oil, diesel) are assumed available for domestic use or imports.

As far as it concerns the natural gas supply, the inland countries which don't have imports, don't have a pipeline infrastructure, and don't have domestic reserves, are not allowed to have any gas imports at any point during the model horizon. The inland countries which have domestic reserves but no future committed natural gas projects can have gas imports from 2025 onwards. Coastal countries which currently don't import natural gas can be supplied after 2025.

The fuel prices for coal, oil products, and gas are based on the World Bank Commodity Markets Outlook [18] [19] [20] and World Energy Outlook 2015 [9]. It is assumed that all countries have the same price as the South Africa's coal (6,000 kcal/kg) until 2015. The period 2016-2025, the price projections of coal are based on the projected growth rates of Australian coal (6,700 kcal/kg). The period 2026-2040, the price projections of coal are based on the growth rates of coal price projections of the World Energy Outlook 2015 for the Low Oil Price Scenario. The crude oil

prices, the period 2010-2025, are based on the World Bank Commodity Markets Outlook and are assumed as import prices. The growth rates of crude oil price projections of the World Energy Outlook 2015 for the Low Oil Price Scenario are implemented the period 2026 onwards. The crude oil domestic prices are assumed to be 5% less than the imported ones the period 2010-2050. For gas import prices, the period 2010-2025 the prices of Europe's natural gas derived from World Bank are assumed, and from 2026 onwards the growth rates of Europe's natural gas imports price projections of the World Energy Outlook 2015 for the Low Oil Price Scenario. The gas domestic prices are assumed to be 5% less than the imported ones the period 2010-2050. As to biomass, it is assumed to be scarce and its prices defined accordingly [21]. The assumed price evolutions for fuels are summarized in Table 4.

Table 4 Fuel Price Projections [9] [18] [19] [20] [21].

USD/GJ (2010 real prices)	2015	2020	2030	2040
Crude oil (USD/bbl)	48	51.2	74.26	90.17
Coal (bituminous,domestic)	1.25	1.17	1.36	1.43
Coal (bituminous,imported,coastal)	1.88	1.75	2.03	2.14
Coal (lignite,domestic)	1.49	1.39	1.62	1.70
Diesel (domestic)	9.99	10.65	15.45	18.76
Diesel (imported,coastal)	10.51	11.21	16.26	19.75
HFO (domestic)	5.89	6.28	9.11	11.06
HFO (imported,coastal)	6.20	6.61	9.59	11.64
Gas (domestic)	6.92	5.91	7.72	9.88
Gas (imported)	7.28	6.22	8.12	10.40
Biomass (scarce)	1.50	1.50	1.50	1.50

4.6 Renewable Resource Potential

Even though Africa has a substantial solar potential, only South Africa (1.5GW) and Algeria (375MW) have done investments towards that direction. The same applies for wind potential, where except the Northern African countries (1,5 GW) and South Africa (1GW), in the rest of the continent this energy source remains unexploited [22]. Especially in countries, such as Somalia, Kenya, Chad, Sudan, with wind speeds more than 9 m/s, wind investments are relatively low [23]. The continent's technically exploitable resource potential from geothermal is about 18 GW and is much smaller than any of the other primary fuel sources. Kenya and Ethiopia are the dominant countries representing 83% of the total capacity [24].

Over the next decades, in order for Africa's rapidly growing electricity demand to get covered, this significantly renewable resource potential should be utilized. Estimates about each country's renewable resource potential: solar, wind, [22] and geothermal [24] [25] can be found in Table 14 of Appendix B. Hydro potential identified in country's reports and other sources [25] [26].

4.7 Electricity generation options

Existing electricity generating capacity

Africa is significantly rich in renewable energy resources, nevertheless the power sector depends on non-renewable power generation options and only around 19% comes from renewables. Although in Central and Eastern Africa the renewable technology shares in the power sector are 65% and 63% in Northern and Southern Africa which are the dominant markets of Africa's electricity demand, their renewable shares are almost around 9% and 18%. The existing power generating capacities by energy source and region until 2015 are summarized in Table 5 [3].

Table 5 Existing power generating capacity by region [3].

Region (MW)	Coal	Diesel/HFO	Gas	Hydro	Solar	Wind	Geothermal	Biomass
Central Africa	0	658	873	2,886	5	0	0	3
Eastern Africa	26	3,210	766	5,930	33	242	691	22
Northern Africa	2,585	4,786	56,330	4,543	411	1,503	0	0
Southern Africa	41,289	2,963	1,375	7,624	1,585	1,010	0	7
Western Africa	68	3,242	13,292	5,025	55	12	0	10

Identified electricity generation projects

The penetration of renewable energy technologies in Africa's future power sector are being highlighted in the identified future power generation projects. The future renewable capacities account for 53% of the total additions, and this is mainly due to hydro in Central (98%), and Eastern (84%) Africa. Northern African countries are expanding significantly their gas industries, only Egypt invests in coal power plants. Southern African countries are mainly investing in coal and gas power plants, especially South Africa (coal - 9740MW, gas - 4009MW) and Zimbabwe (coal - 5100MW), but their investments especially in hydro and wind represent 47% of the planned addition.

The summary of future power generation projects, committed or under consideration, by energy source and region are summarized in Table 6 [3]. Only the projects which are committed are forced to be used into the future energy system in the model, those which are under consideration are selected by the optimization process.

Table 6 Identified power generating capacity additions by region [3].

Region (MW)	Coal	Diesel/HFO	Gas	Hydro	Solar	Wind	Geothermal	Biomass
Central Africa	500	6	471	48,034	212	0	0	0
Eastern Africa	3,176	757	2,312	30,361	561	1,086	1,310	64
Northern Africa	6,270	20	41,395	281	976	757	0	0
Southern Africa	21,064	868	7,047	21,231	2,167	2,359	0	72
Western Africa	7,390	106	17,524	10,971	839	420	0	45

4.8 Local transmission and distribution technologies

Most African countries have unreliable grid networks with high losses, almost double than the world average, and high electricity tariffs [5]. Each country's national grid is represented in the model by transmission and distribution technologies, defined by losses based on country's historical data. The off-grid technologies, since they don't require transmission and distribution infrastructure, they don't have any losses and costs. In this study the transmission losses remain constant over time. Different distribution losses and costs are assigned in each country and in each of the three demand categories (Table 16, Appendix C). In distribution technologies efficiency improvements are being made through the years. Table 7 shows the costs of transmission and distribution infrastructure [16].

Table 7 Assumptions for Transmission and Distribution infrastructure [16].

Trans. & Dist. Cost	
	USD/KW
Transmission (all sectors)	364.97
Others	840.4
Urban	2433.3
Rural	4233.6

4.9 International Trade Links

A cross-border interconnected grid is one of the main driving factors that positively affect the development process among the trading countries. It promotes security of electricity supply, in case the country faces generation shortages, and gives incentives for lower electricity tariffs. Countries which have low costs of generating electricity can provide low electricity prices to their customers and concurrently export to other countries to increase their profits. Additionally, the costs for the importing countries will be lower, rather than generating more expensive electricity to cover their energy needs. However, the current grid infrastructure among the African countries

is limited and thus can't support extensive trade. The following categories of international trade technologies are considered in the model for the different scenarios (Table 17, Appendix C)[1].

- ◆ Existing capacities: interconnections which are available from the base year of the model or before, and operate throughout the model period. They are included in the Reference scenario.
- ◆ Future capacities:
 - Committed: interconnections which are or will be constructed throughout the model period and their operation year has been identified. They are included in the Reference scenario.
 - Planned: interconnections which will be constructed. Their operation year has been identified, but uncertainties in the implementation process in some of the infrastructure projects exclude their total capacities to be taken into account in the Reference scenario. If the interconnections have been granted financing, they are included in the Reference scenario. Otherwise they are only considered in the Enhanced scenario.
 - Assumed: future uncommitted interconnections projects which allow trade links among neighboring countries from 2025 onwards, and not included in the existing and committed projects. These capacities are included only in the Enhanced scenario.

5. Results

5.1 Reference scenario

The significant increase of population, and electricity access of Africa in the future increase on a large scale the energy needs of the continent and thus investments required. The total installed capacity of the African countries was 170GW in 2015, doubles to 364GW in 2030, at which point the electricity access in the continent is 100%, and further increases to 785GW in 2050 (Figure 7). In order for Sub-Saharan Africa to be electrified, its total installed capacity needs to be increased from 96GW in 2015, to 238GW in 2030, and 595GW in 2050.

In 2015, the Northern African power pool (74GW) and Southern African power pool (58GW) had the biggest share of the continent's installed capacity. The Central African power pool (5GW), Eastern African power pool (11GW), and Western African power pool (22GW) increase their installed capacities in 2030 approximately four times compared to 2015 levels, and twelve times by 2050 to cover their energy needs. The Western African power pool's (228GW) installed capacity, followed by Northern African power pool (189GW), represent most of the continent's installed capacity in 2050. The enormous increase in the total capacity of Western African power pool is due to the low electrification rates and high population which identify the power pool. This is also reflected by the increase of Nigeria's installed capacity from 14GW in 2015 to 109GW in 2050. In Northern African power pool, the total installed capacity of Egypt 33GW in 2015, and 112GW in 2050, corresponds to most of power pool's investments. Eastern African power pool followed by Central African power pool have the biggest increase in their installed capacity, from 11GW, and 5GW in 2015, to 151GW, and 61GW in 2050, correspondingly. The installed capacities of Ethiopia,

and Sudan, 2.32GW, and 2.6GW in 2015, and 35.56GW, and 37GW in 2050, represent the biggest shares of the installed capacities in the Eastern African power pool.

In 2050, hydro and CSP with storage technologies represent approximately 70% of the continent's total installed capacity. In Central African power pool, the investments in hydro represent around 75% of each year's total installed capacity with the share of CSP with storage technologies to be increased at the future. Also, several countries, like Egypt, Morocco, Kenya, and South Africa, have started to harness their wind resources, and continent's wind capacity increases from 7GW in 2020 to 42GW in 2050. Investments in geothermal technology are only held in Eastern African power pool, and mainly by Kenya and Ethiopia, 8.2GW, and 2.5GW in 2040. Nuclear technology investments held in Northern African power pool and Southern African power pool only by South Africa and Egypt.

In the continent, gas investments are mainly in Northern African power pool, with Algeria, and Egypt, 17GW, and 30GW, out of 57GW in total in 2030, to be responsible for most of this share. In 2030, Western African power pool, and mainly Nigeria (11.6GW) constitutes most of the continent's gas installed capacity. The installed capacity of coal in Southern African power pool decreases throughout the years, from 42GW in 2015 to 7GW in 2050, since South Africa's coal capacity decreases from approximately 39GW in 2015 to 3GW in 2050. Western African power pool increases its coal capacity from 0.07GW in 2015 to 19GW in 2050, mainly due to Nigeria's high energy needs 510TWh in 2050. Nigeria shift its investments apart from renewable energy technologies to coal as well, with country's coal installed capacity to reach 11GW in 2050.

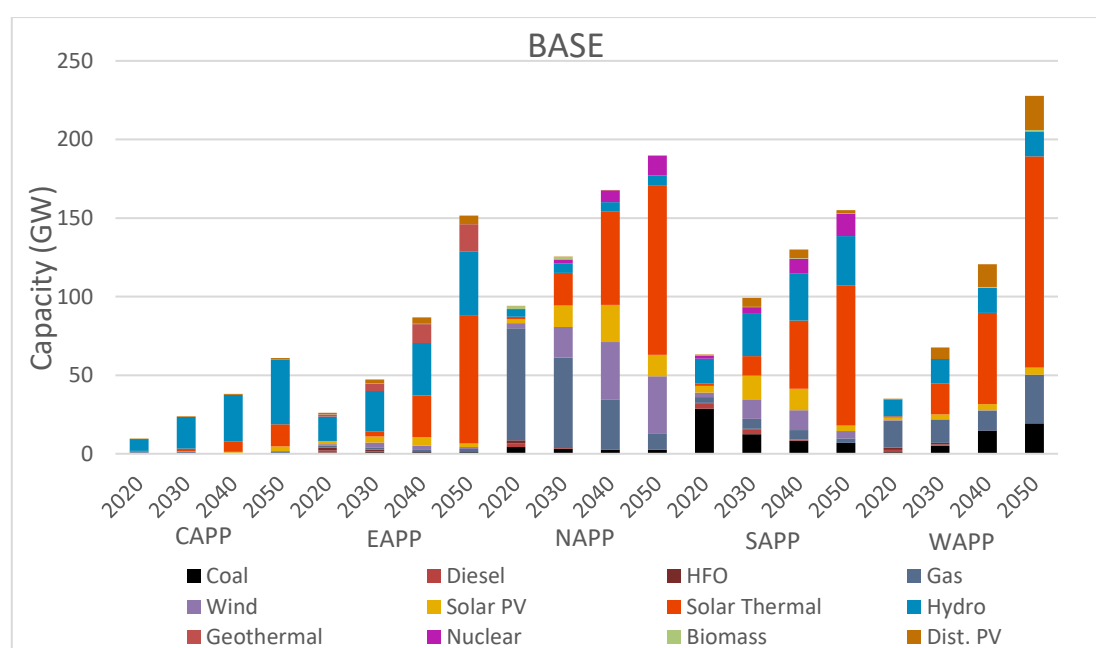


Figure 7 Total installed capacity per power pool in the Reference scenario the period 2020-2050.

The electricity generation in the continent increases from 734TWh in 2015 to 1498TWh in 2030, and to 3854TWh in 2050 (Figure 8). The Northern (382TWh) and Southern (318TWh) African power pools constitute most of the continent's electricity generation in 2020. However, the

Western (1110TWh) African power pool, and the Southern (820TWh) African power pool are responsible for most of the continent's electricity generation in 2050. Western and Eastern African power pools present the largest increase in production of electricity throughout the years.

The penetration of renewable energy technologies, such as hydro, solar and wind, in the electricity generation mix of the countries increases through the years. Countries such as, South Africa and Morocco, where coal represented 90% and 54% of their total electricity generation in 2014 [6], at the future they exploit their renewable energy potential, and these shares shift to approximately 1% (Figure 9). Also, the electricity generation mix of Niger in 2014 was based on fossil fuels (coal, oil) but through the years, hydro and solar technologies substitute these sources. The electricity generation of the Central African power pool at the future is based on hydro, representing approximately 82% of its total generation, and the rest comes mainly from solar thermal technologies. The implementation of the Inga III (4.8GW) and Grand Inga (39GW) hydropower projects in Democratic Rep. of Congo the upcoming years could exploit the enormous hydro potential of the country.

The Central African power pool constitutes most of the continent's electricity exports the period 2020-2050, 12TWh in 2020, and 42TWh in 2050. An increase which is mainly due to Democratic Rep. of Congo hydropower potential since the other countries like Cameroon, and Congo have a small share of the power pool's net imports. Democratic Rep. of Congo exports most of its electricity to Namibia and South Africa through the years. Southern African power pool presents most of the continent's net imports, where Eastern and Northern African power pool display a relative increase of their net imports through the years. The share of each country's electricity generation mix in 2050 is presented in Figure 9, and the trades among the African countries (2020,2030,2040,2050) can be found in Figure 18 - Figure 21 of Appendix D.

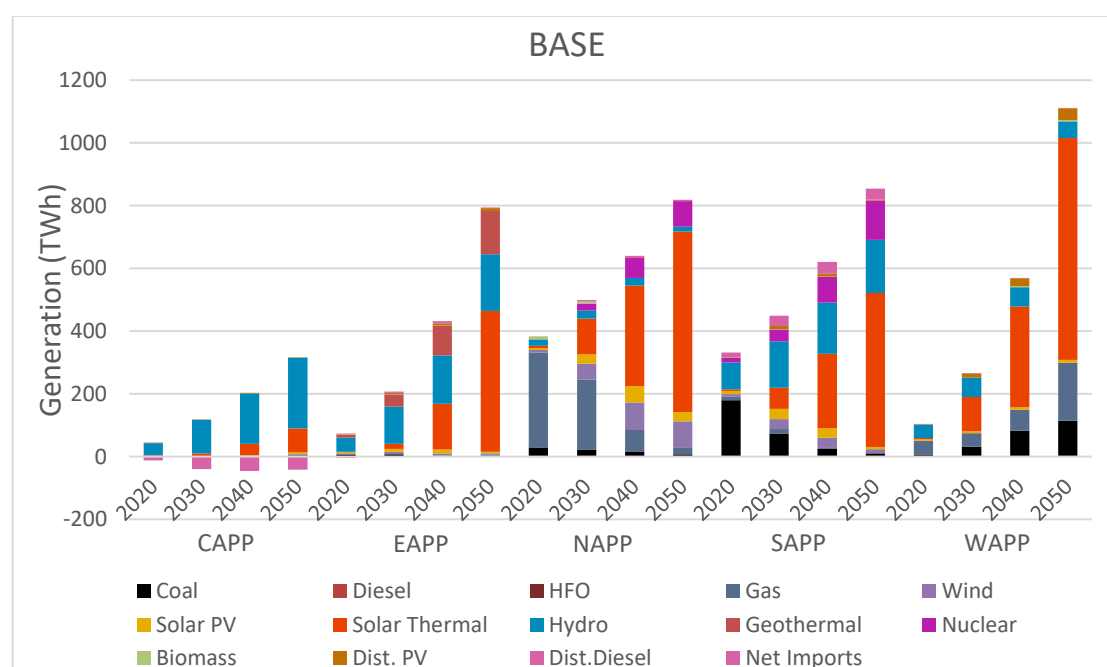


Figure 8 Electricity generation mix per power pool in the Reference scenario the period 2020-2050.

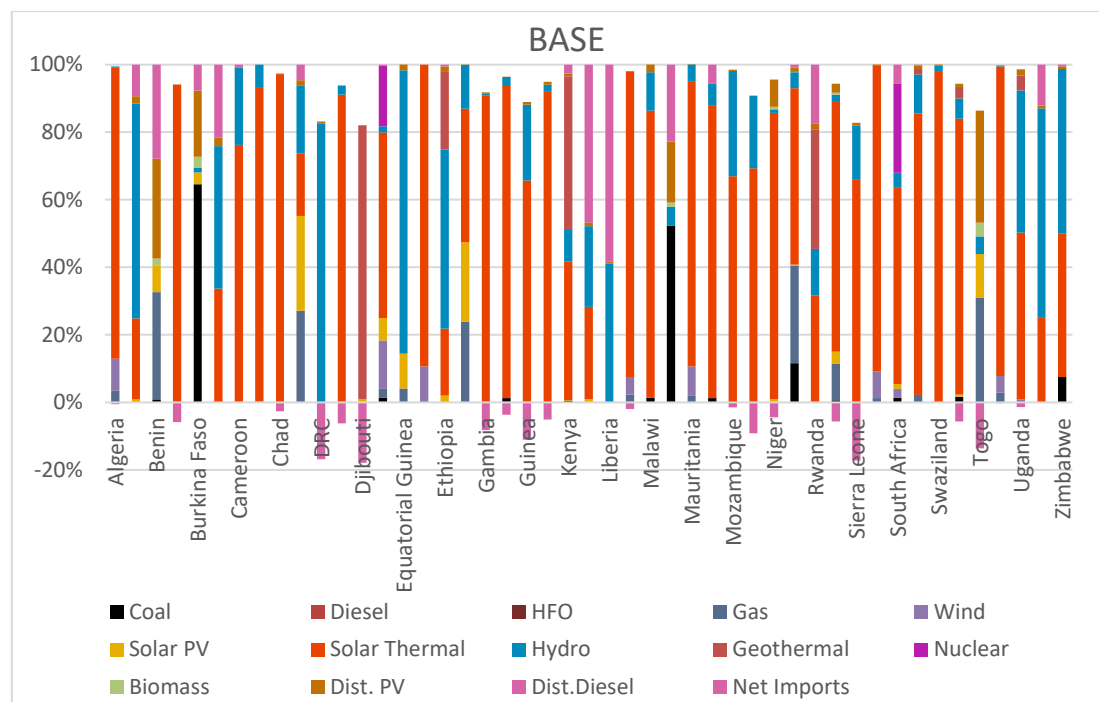


Figure 9 Share of electricity generation mix by country in the Reference scenario in 2050.

5.2 Enhanced scenario

In this scenario, the total installed capacity of the African countries 170GW in 2015, increases to 361GW in 2030, and to 757GW in 2050 (Figure 10). The potential cross-border electricity interconnections project among neighboring countries are considered in the model after 2025. The total installed capacity in Sub-Saharan Africa from 96GW in 2015, increases to 241GW in 2030, and to 578GW in 2050. The total installed capacities of Central, Eastern, Southern, and Western African power pool are 28GW, 53GW, 93GW, and 66GW in 2030, with these levels to increase to 72GW, 145GW, 141GW, and 220GW in 2050. The Central and Eastern African power pool represent the highest increase in their installed capacities throughout the years, fifteen and thirteen times correspondingly in 2050, compare to 2015 levels. The Western African power pool (220GW), and Northern African power pool (179GW) in 2050 represent most of the continent's total installed capacity. Nigeria, and Egypt installed capacity is 100GW correspondingly in 2050, representing approximately 26% of continent's total capacity. The installed capacity of Democratic Rep. of Congo 18GW in 2030, and 47GW in 2050, constitutes most of Central African power pool power system.

The installed capacity of fossil fuels in the continent throughout the years is decreasing, 73GW in 2050 compare to 143GW in 2020. Western African power pool's coal capacity increases from 1GW in 2020 to 31GW in 2050, and constitutes most of the continent's share the end year. This increase is mainly due to the shift of Nigeria's energy mix from gas 13GW in 2020, and 12GW in 2050, to coal 0.3GW in 2020, and 25GW in 2050. In the continent, gas investments are decreasing, especially in Northern African power pool, apart from Eastern African power pool, from 71GW in 2020 to 12GW in 2050, due to the shift of Egypt and Algeria to technologies such as solar, hydro, and wind.

Hydro and CSP with storage technologies represent approximately 74% of the continent's total installed capacity in 2050. The installed capacity of geothermal in Eastern African power pool

increases from 1GW in 2020 to 18GW in 2050. This share increases mainly due to Ethiopia, and Kenya, 5GW, and 10GW in 2050 correspondingly, since these countries have started to harness their geothermal resources at a higher level. The installed capacity of wind potential in the continent increases from 7GW in 2020 to 23GW in 2050, mainly due to its harnessing by Egypt (15GW), Algeria(3GW) and South Africa(3GW) in 2050. South Africa and Egypt constitute for continent's total nuclear installed capacity.

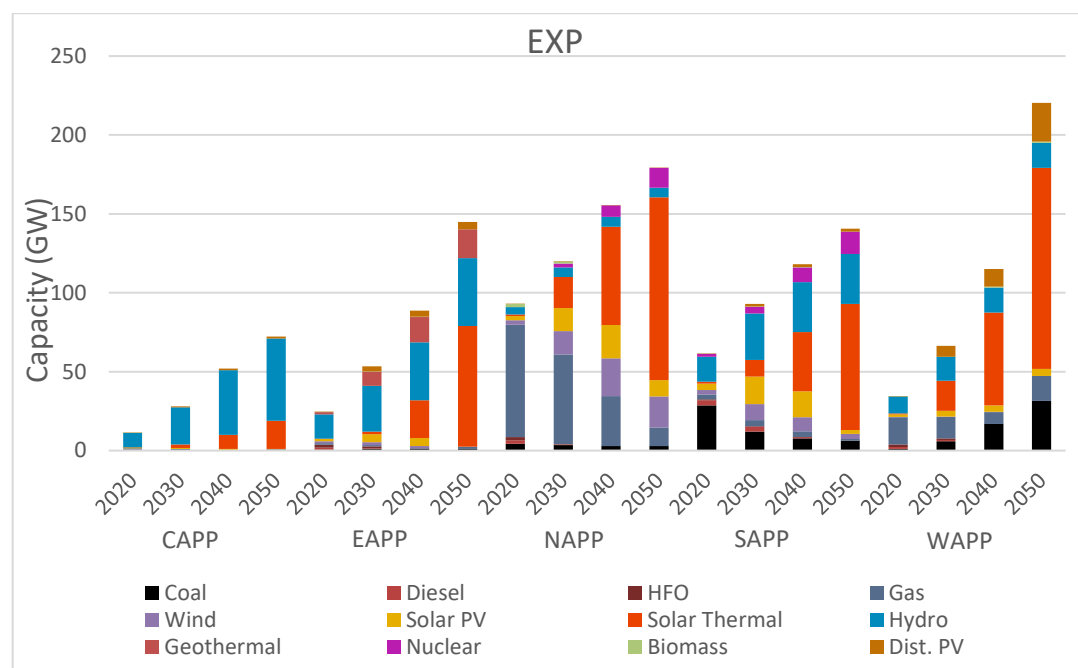


Figure 10 Total installed capacity per power pool in the Enhanced scenario the period 2020-2050.

The electricity generation in the continent from 734TWh in 2015, increases to 1509TWh in 2030, and to 3864TWh in 2050 (Figure 11). Most of the continent's electricity generation in 2050 comes by Western African power pool (1085GW), and Northern African power pool (852TWh). However, Eastern, and Western African power pool represent the largest increase on their generation, approximately eight and ten times correspondingly compare to 2020 values.

The generation from fossil fuels in 2020 constitute approximately 61% of continent's electricity production, with this share to drop to 9% in 2050. Only Western African power pool increases its coal based electricity generation throughout the years. This is mainly due to Nigeria's coal electricity generation increase from 2TWh in 2020 to 160TWh in 2050, since Ghana's coal electricity generation decreases from 4TWh in 2020 to 2.5TWh in 2050. The coal electricity generation from Nigeria throughout the model period represents approximately 86% of the power pool's and 75% of the continent's coal electricity generation. Countries such as Egypt, and Morocco which represent most of the Northern African power pool coal electricity generation in 2020, 14TWh and 14TWh correspondingly, shift to renewable energy technologies at the future and these values drop to 7TWh and 2TWh respectively.

South Africa starts to harness its renewable potential (hydro, solar, wind) at the future, and the renewable's share in country's electricity generation mix present an enormous increase from 12% in 2020 to 70% in 2050. In Northern African power pool is highlighted the penetration of wind technologies in the electricity generation mix of the continent. Countries such as Algeria, and Egypt produce 9TWh, and 34TWh correspondingly by this energy source in 2050.

Central African power pool represents most of the continent's exports throughout the years with an enormous increase from 21TWh in 2020 to 95TWh in 2050. Most of this increase, is based on the increase of hydro electricity generation from 48TWh in 2020 to 290TWh in 2050, as well as on CSP with storage 98TWh in 2050. Moreover, Eastern African power pool represents exports until 2040 mainly due to Ethiopia and Sudan. In 2020, Ethiopia constitutes the largest exporter in the continent 25TWh, by harnessing its hydro and geothermal potential. Most of the country's electricity exports go to Sudan 23TWh, where Sudan export it as well to Egypt 17TWh. South Africa is responsible for most of the continent's electricity imports 53TWh in 2020 representing approximately 28% of its total electricity generation. In 2050, Democratic Rep. of Congo represent most of Africa's electricity exports 106TWh based on its large hydro potential, with South Africa 24TWh and Namibia 24TWh to import most of this produced electricity.

The expansion of the cross-border electricity interconnections among African countries lead several countries, such as Central African Rep., Eritrea, Gabon, Equatorial Guinea, Mauritania, and Malawi to be included in the continent's electricity trading scheme. These interconnections contribute as well to the continent's sustainable development through harnessing each country's energy resources. The share of each country's electricity generation mix in 2050 is presented in Figure 12, and the trades among the African countries can be found in Figure 22 - Figure 25 of Appendix D.

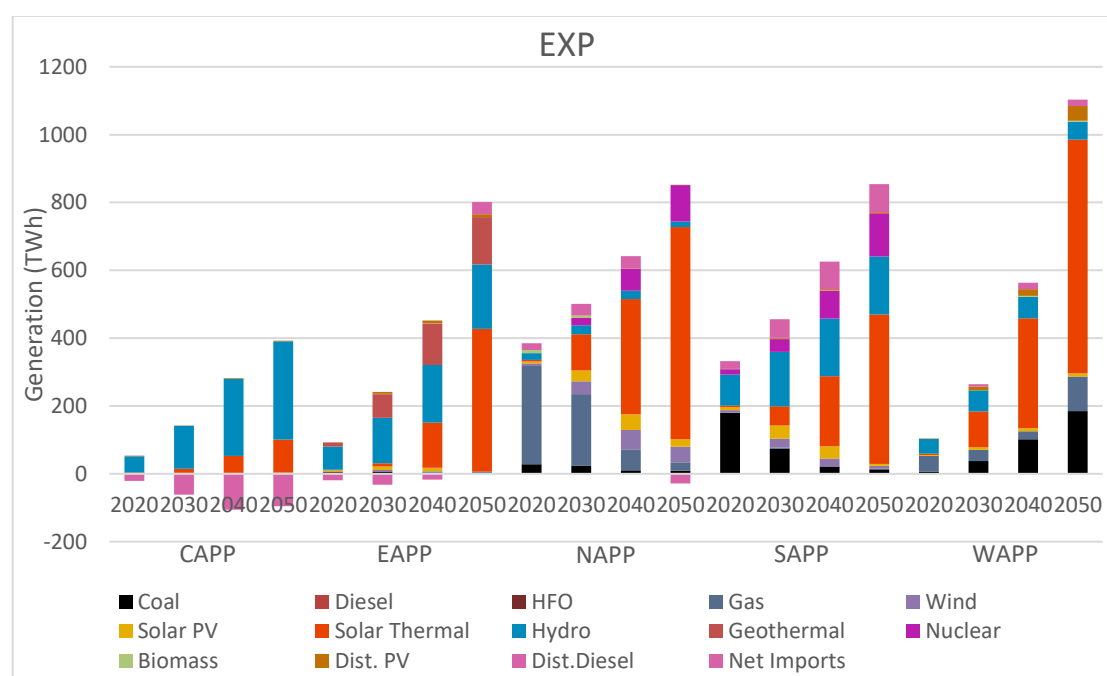


Figure 11 Electricity generation mix per power pool in the Enhanced scenario the period 2020-2050.

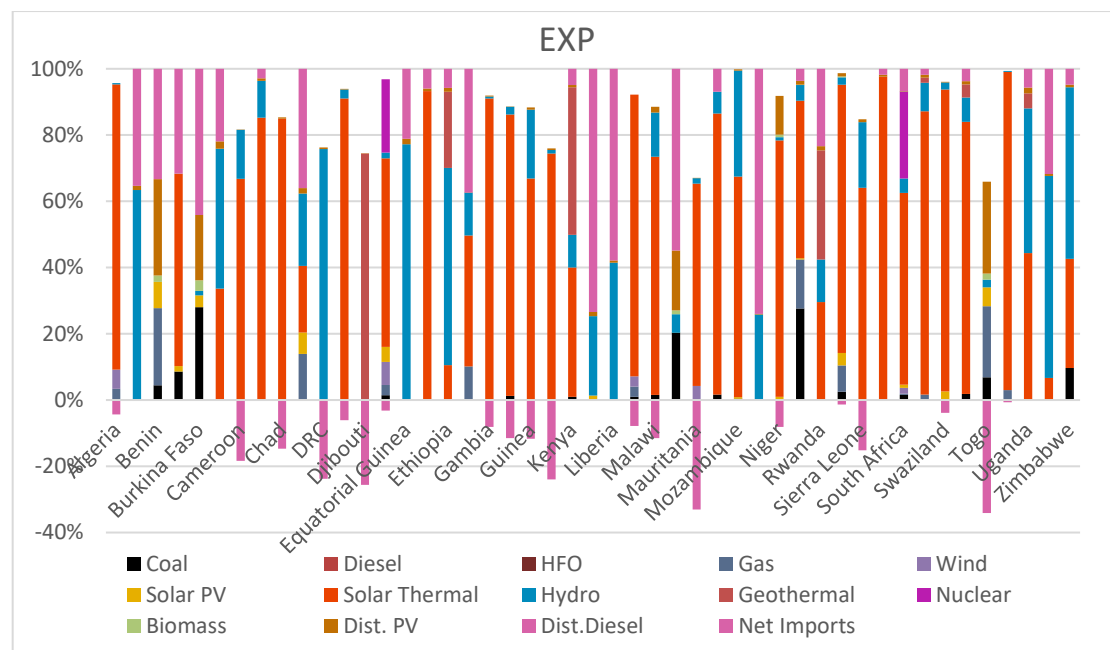


Figure 12 Share of electricity generation mix by country in the Enhanced scenario in 2050.

5.3 Scenario comparison

This thesis compares the two aforementioned scenarios based on the cross-border electricity interconnection projects among the African countries. The total installed capacity of Africa in both scenarios was 170GW in 2015. In 2030, the total installed capacity in the Reference scenario is 364GW compared to the Enhanced 361GW. In 2050, these values increase to 785GW, and 757GW correspondingly in each scenario. Between the two scenarios, the Central African power pool represents the highest increase of its installed capacity through the years, in the Enhanced more than in the Reference scenario, mainly due to the increase of Democratic Rep. of Congo hydro installed capacity for more electricity exports. In contrast, the Southern and Northern African power pool installed capacities represents the highest decrease between the two scenarios mainly due to South Africa's and Egypt's decrease of their installed capacities, and increase of their net imports.

The total electricity generation in the continent in both scenarios was 734TWh in 2015. In 2030, the total electricity generation in the Reference scenario is 1498TWh and in the Enhanced one is 1509TWh, where in 2050 is 3855TWh, and 3864TWh correspondingly. The installed capacity of the continent between the two scenarios decreases throughout the years. Nevertheless, in the Enhanced scenario the electricity generation increases. This is mainly due to the expansion of the trading scheme among the countries since further losses are included, and more efficient technologies are being used.

The Central African power pool constitutes most of Africa's electricity exports through harnessing of its hydro and solar potential, with Democratic Rep. of Congo 42TWh in Reference, compared to 106TWh in Enhanced in 2050, to be the largest electricity exporter in the continent. In the Enhanced scenario, most of Democratic Rep. of Congo electricity is exported to Namibia (25TWh), South Africa (24TWh) and Zambia (19TWh) in 2050, where Namibia exports 9TWh as well to South Africa in 2050 helping the country to cover its energy needs. The effects of the

expanded grid electricity interconnections among the African countries is more obvious in the Southern African power pool. The net imports in this power pool are approximately doubled throughout the years, and substitute the electricity generation based mainly on csp with storage and less on wind technologies. In the Enhanced scenario, the role of Eastern African power pool in the continent's trading scheme is highlighted, where by harnessing its hydro and geothermal potential, represent a big electricity exporter in the continent until 2040. Ethiopia even that most of its electricity is exported to Sudan 45TWh in 2030, Sudan exports as well 32TWh to Egypt that year. Sudan covers approximately 13% of Egypt's electricity demand in 2030 through its electricity exports, and constitutes Egypt's largest importer until 2040. This shift in the electricity generation mix of Egypt, which is the largest electricity consumer in the Northern African power pool, affects the exploitation of the power pool's wind potential. The differences in the electricity generation mix in each power pool between the two scenarios are presented in the Figure 13 below.

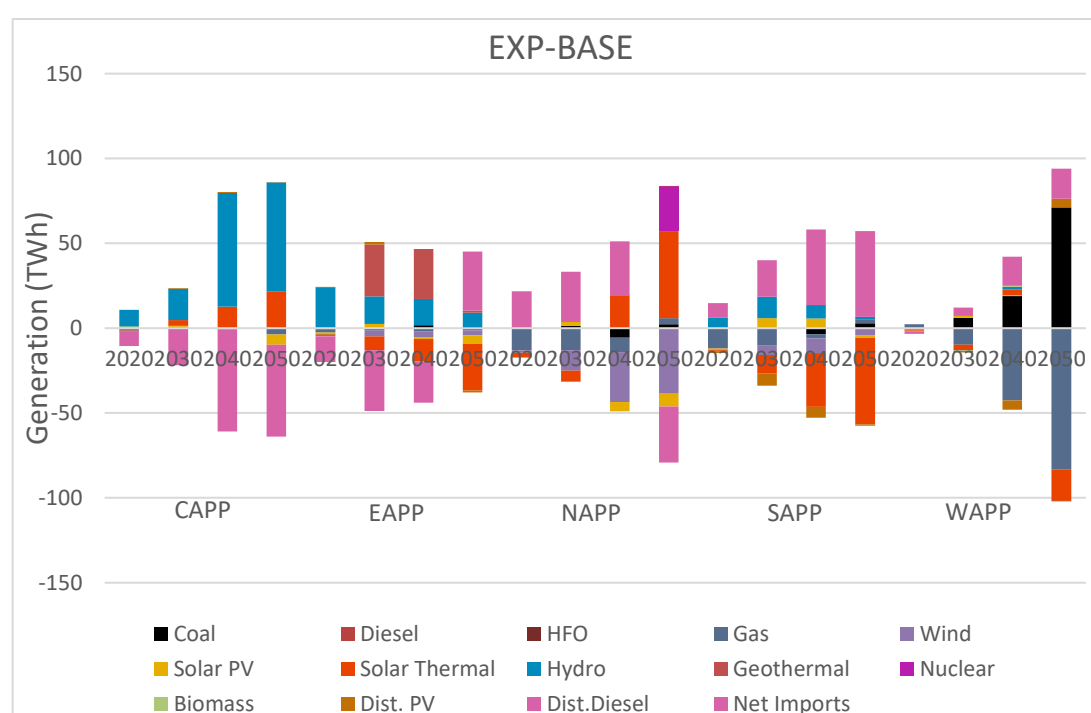


Figure 13 Comparison of the electricity generation mix per power pool the period 2020-2050: Enhanced – Reference.

The penetration of renewable energy technologies in the power sector of the continent is justified in both Reference and Enhanced scenarios, where the decrease of fossil fuels shares in the electricity generation mix in the first scenario from 64% in 2020 drops to 9% in 2050, and in the second one from 61% in 2020 to 9% in 2050. These shares would have dropped at a higher scale in 2050, if the Western African power pool, and mainly Nigeria's fossil fuel electricity generation which represents approximately 85% of continent's fossil fuel electricity generation in 2050, had been substituted by other renewable sources. The enormous increase of Nigeria's electricity demand the upcoming decades leads the country to exploit its significant identified recoverable fossil fuel reserves. Also, even though South Africa's net imports increase between the two scenarios, the increase of the fossil fuel share in country's total electricity generation from

approximately 1% in the Reference to 2% in the Enhanced scenario after 2040, constitute to most of the carbon dioxide emissions increase in that specific power pool.

Hydro, and csp with storage technologies are in both scenarios the main power generation technologies of the continent and pillars of its sustainable development, representing around 65% of its total electricity generation. Eastern African power pool has increased its electricity generation based on geothermal potential approximately by 20% in the Enhanced scenario throughout the years. The levels of biomass exploitation as an energy source for electricity generation in Southern and Western African power pool remain constant in the two scenarios. The carbon dioxide emissions per power pool as well as the total emissions of the continent between the two scenarios, are presented in Table 8 and Figure 14 correspondingly. The carbon dioxide emissions are higher in the Enhanced scenario compared to the Reference one the period 2040-2050, which does not correspond with the comparison of the two scenarios in the period up to 2040. Additional sensitivity analysis is needed to fully understand the dynamics of the model.

Table 8 Carbon dioxide emissions per power pool.

	CO ₂ emissions (Mt)							
	BASE scenario				EXP scenario			
	2020	2030	2040	2050	2020	2030	2040	2050
CAPP	2.20	0.21	0.94	2.53	1.27	0.54	0.57	1.11
EAPP	5.79	6.49	2.32	4.37	4.52	5.90	3.13	4.48
NAPP	142.00	103.97	40.09	14.56	136.95	100.35	32.09	17.53
SAPP	160.99	69.38	22.68	9.22	156.32	65.54	18.68	12.54
WAPP	22.75	43.39	96.87	168.80	23.00	45.05	97.29	199.33
Total CO₂	333.74	223.44	162.91	199.48	322.06	217.38	151.76	234.99

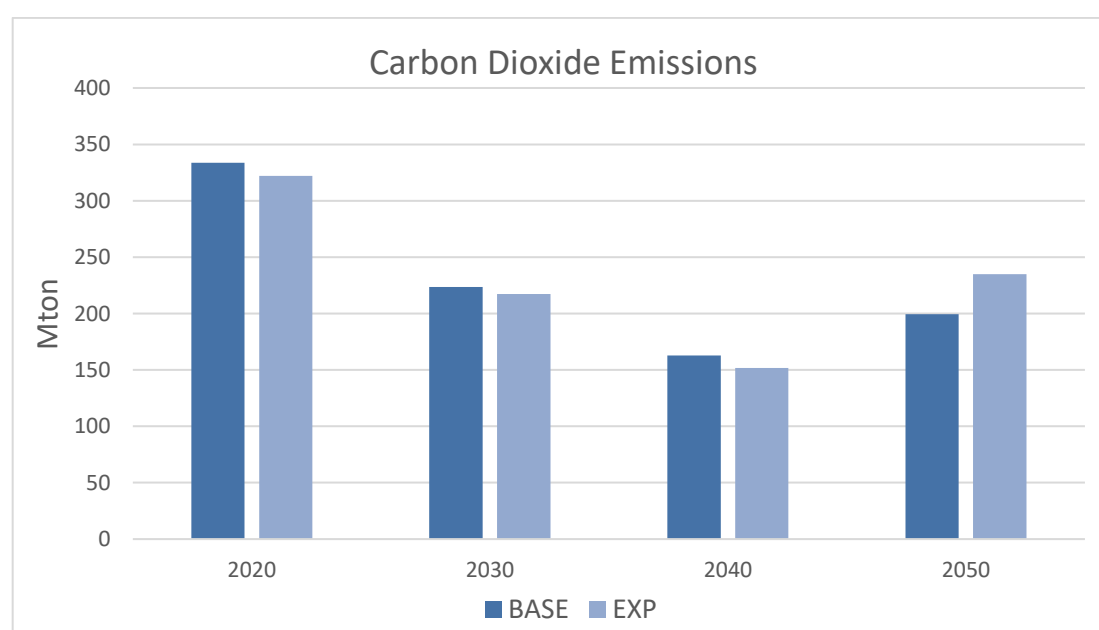


Figure 14 Comparison of the carbon dioxide emissions between the scenarios the period 2020-2050.

The fuel costs of the continent throughout the years are increasing as the African countries boost their electricity demands. In the period 2016-2020, the fuels costs in the Reference scenario are 131 billion USD compare to 125 billion USD in the Enhanced scenario, and the period 2046-2050 are 53 billion USD and 47 billion USD correspondingly (Figure 15). The expansion of the electricity trading system among the countries, especially with the potential trading projects to be held after 2025, lead the continent to have less fuel costs between the two scenarios, from 5% decrease in fuel costs in 2016-2020, this share to increase to 12% in 2046-2050. The exploitation of each country's renewable potential for electricity production in the Enhanced scenario increased the net imports of the countries, and reduced the fuel costs. Also, the shift from gas to coal electricity generation in the Western African power pool between the two scenarios affected the continent's fuel costs.

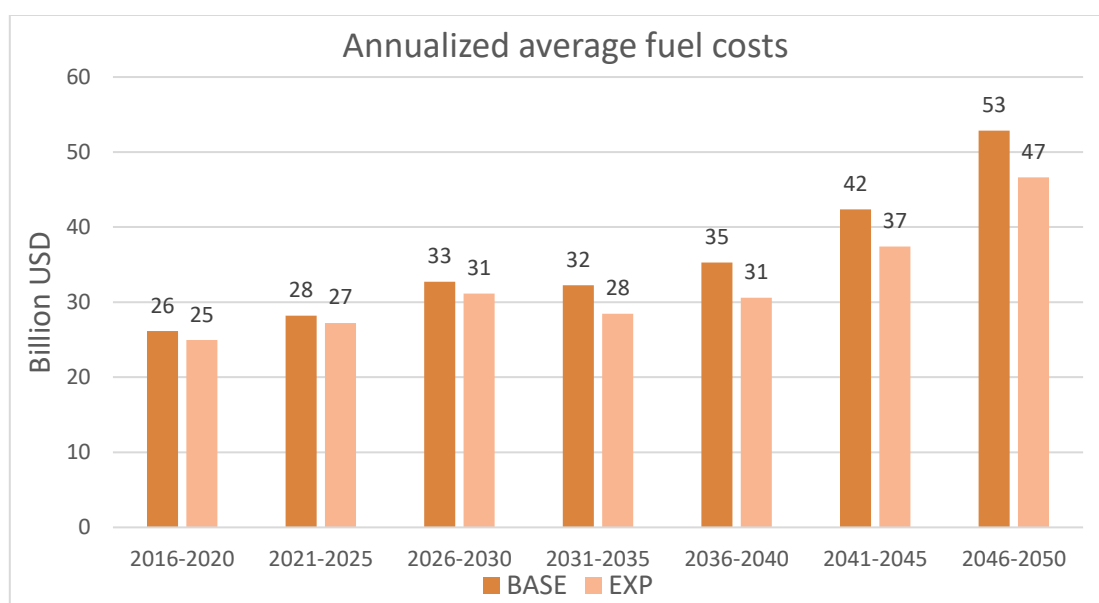


Figure 15 Comparison of the annualized average of the total fuel costs between the scenarios the period 2016-2050.

Africa in order to be fully electrified in 2030 and be able to cover its electricity demands, investments of approximately 808 billion USD are needed between 2016 and 2030 in the Reference scenario, compare to 819 billion USD in the Enhanced scenario. The annualized average of the total investment costs between the two scenarios are represented in Figure 16. The total investment costs the period 2010-2050, in the Reference scenario are 3332 billion USD and in the Enhanced 3302 billion USD. The electricity investment costs the period 2026-2035, in the Enhanced scenario are 16billion USD more than in the Reference scenario. This increase is mainly due to the investments of the cross-border electricity interconnections projects among the countries which take place after 2025, since the installed capacity of the continent in the Reference scenario (364GW) is higher than the Enhanced (361GW) in 2030. However, in the long term the total investment costs in the Reference scenario are higher than in the Enhanced one. In the Enhanced scenario, the increase of Northern and Southern African power pool net imports and decrease of their installed capacities by 5% and 9% correspondingly from 2030 onwards, contribute to lower the investment costs and highlight the role of the trading scheme in this scenario. The period 2016-2020, the renewable investments constitute approximately 57% of the total investments that

period, with this share to increase to an average around 75% in the following years in both scenarios.

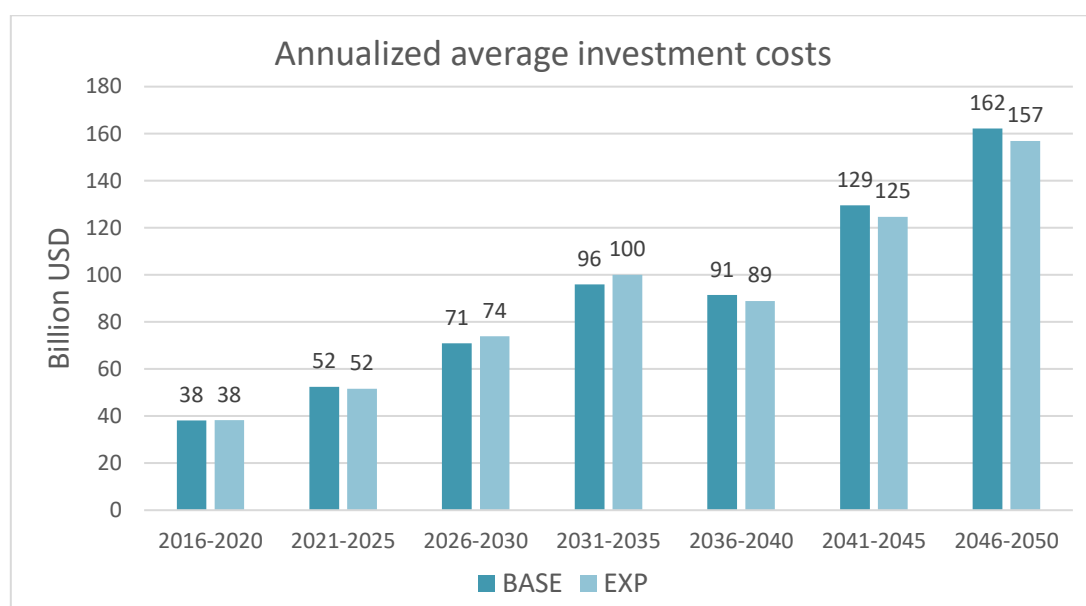


Figure 16 Comparison of the annualized average investments costs between the scenarios the period 2016-2050.

The reduction of continent's investment and fuel costs between the two scenarios for generating electricity could significantly improve the overall energy supply in Africa. Strategic investments towards the expansion of cross-border electricity interconnections projects among the African countries estimate lower electricity tariffs for the continent. The electricity price of the continent throughout the years represents a decrease of approximately 5% between the Reference and the Enhanced scenario (Figure 17). The electricity price of the system is 0.046 USD/kWh in 2015, increases to 0.06 USD/kWh in the Reference scenario compare to 0.057 USD/kWh in the Enhanced scenario in 2030, and grows to 0.064 USD/kWh and 0.062 USD/kWh correspondingly in 2050. The annualized electricity prices in each country vary, mainly due to differences in their existing power system's infrastructure and availability of energy resources.

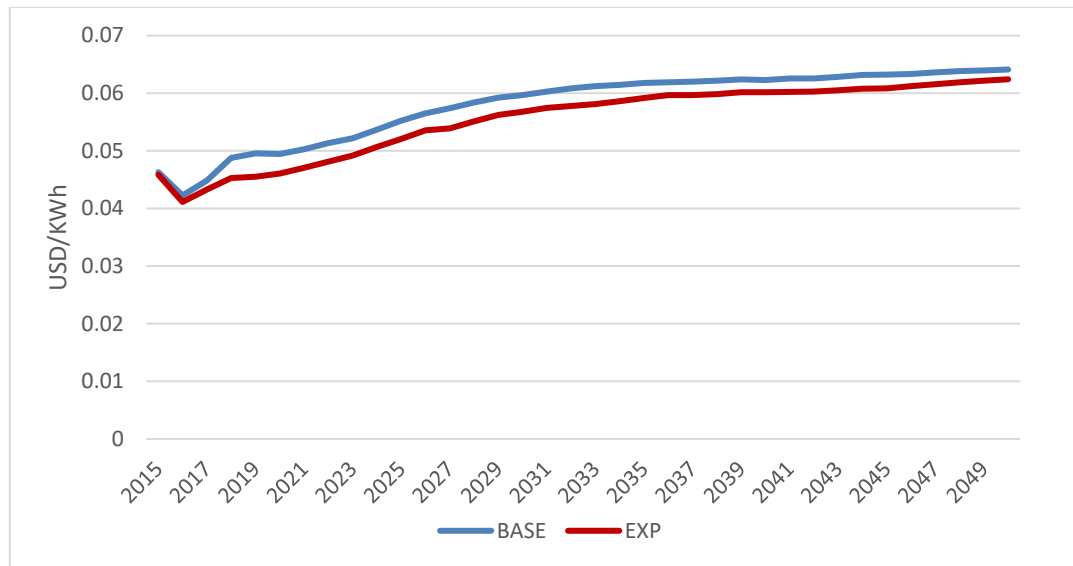


Figure 17 Comparison of the total annualized electricity price between the scenarios the period 2010-2050.

6. Conclusions

This thesis indicates that the associated investment and fuel costs which are needed to fully electrify Africa until 2030, as well as the continent to cover its electricity demands the period 2010-2050, are significantly affected by an expanded electricity trading scheme. Between the two scenarios, the total costs are represented a total average decrease of 3%. Investments of approximately 808 billion USD are needed between 2016-2030 in the Reference scenario, compare to 819 billion USD in the Enhanced scenario. In the upcoming decades, the renewable investments will constitute approximately an average of 75% of the total investments. The total fuel costs have decreased approximately by 8% between the two scenarios the period 2010-2050. Less investment and fuel costs contribute to reduce the electricity price of the continent throughout the years approximately by 5% between the Reference and the Enhanced scenario.

The total installed capacity in Africa 170 GW in 2015, in the Reference scenario increases to 785 GW compare to 757 GW in the Enhanced scenario in 2050. The expansion of the continent's trading scheme allows the system to harness each country's energy resource potential and use the most efficient technologies. This lead to an increase of the total electricity generation from 3855 TWh in the Reference to 3864 TWh in the Enhance scenario in 2050. In the Enhanced scenario the role of the Central African power pool is being highlighted as the largest electricity exporter in the continent by harnessing its significant hydro potential mainly by Democratic Rep. of Congo. The implementation of a project such as Grand Inga would exploit country's hydro potential and become a key player in the continent's trading scheme.

The penetration of renewable energy technologies in the power sector of the continent is highlighted in both Reference and Enhanced scenarios the upcoming decades. The shares of fossil fuels in the electricity generation mix of the continent are decreasing throughout the years. That share from 65% in 2013 drops to 9% in 2050 [8]. An increase on the renewable capital investments by approximately 87 billion USD the period 2021-2025 compared to 2016-2020, around 25% increase of the total investment in generation infrastructure, lead to a decrease of approximately

20% of continent's carbon dioxide emissions. The total carbon dioxide emissions in both scenarios are decreasing through the years, contribute to a more environmental friendly power generation infrastructure which the African leaders pursue to achieve at the future. In 2050, the total carbon dioxide emissions of the continent (199Mt) based on its electricity generation will be less than the total carbon dioxide emissions of Spain (217Mt with emissions removals from land use, land-use change and forestry) in 2013 [27]. As a conclusion, the expansion of the cross-border electricity interconnections projects among the African countries as well as the promotion of renewable generation will have a key role in forming Africa's power sector and contribute to its sustainability the upcoming decades.

7. Limitations-Future work

In this section, the limitations, and the areas for future work, which identified in this thesis and will help to have a broader understanding and a more detailed representation of Africa's electricity system, are highlighted. In this research, it has been assumed that the current unelectrified population of Africa will reach a specific electricity consumption level in 2030 and 2050, without taking into consideration the relationship between the economic growth and electricity consumption of each country [28]. Also, the carbon dioxide emissions are higher in the Enhanced scenario compared to the Reference one the period 2040-2050, which does not correspond with the comparison of the two scenarios in the period up to 2040. Additional sensitivity analysis is needed to fully understand the dynamics of the model. The future research can be focused on improving national specific data (such as electricity demands, resource potentials), analyze explicitly the security and trade by country, investigate the effects of different transmission costs and losses, and developing regional integrated greenhouse gas mitigation cost curves and emissions trade. Also, calculating power supply costs by country and region, understanding the impact of local or technology constraints (such as hydro generation under a changing climate), and developing deep analysis of potential intermittency and transmission constraints related to renewable energy technologies deployment, could lead to better encapsulate the transition of the real world. Moreover, the temporal resolution in the model was limited and the load curve should have disaggregated into more levels (seasons, days) to capture better the variation in load and availability of some of the renewable resources as well as the trades. This limitation was mainly due to the few available national load curves as well as the increase in the computational time of the model. Furthermore, a Monte Carlo technique should be used to understand the influence of the parametric uncertainty and improve the reliability of the model as well as a sensitivity analysis. The two scenarios assessed in this thesis are a starting point and more scenarios should be examined to encapsulate the real-world transition. In future research efforts, several scenarios can be conducted related to GHG constraints, fuel prices, technology cost assumptions, consumption growth rates, as well as many others, so that uncertainties of energy modeling in future changes could be minimized [29]. The open source (OSeMOSYS) toolkit which is used in this thesis provide an open access to the public and therefore future work and formulation of the model beyond the scope of this thesis can be attempted.

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Appendix A: Detailed Demand Data

Table 9 Final electricity demands in each country (TWh).

Region	2010	2015	2020	2030	2040	2050
Africa	540.4	649.8	815.4	1331.0	2143.1	3416.0
Algeria	33.5	47.4	57.8	76.3	97.7	125.1
Angola	4.7	5.6	6.9	13.6	25.6	43.2
Benin	0.8	1.2	1.8	5.2	11.3	21.5
Botswana	3.1	3.6	4.2	5.8	8.3	11.6
Burkina Faso	0.8	1.3	2.1	7.1	16.4	31.4
Burundi	0.2	0.3	0.7	3.3	8.2	15.8
Cameroon	4.9	6.2	8.8	17.8	32.6	56.0
Central African Rep.	0.1	0.2	0.3	1.3	2.9	5.0
Chad	0.2	0.2	0.5	3.5	8.9	16.6
Congo	0.5	1.0	1.4	3.1	5.9	10.3
Congo, Dem. Rep.	6.3	9.1	13.8	37.5	78.6	140.9
Cote d'Ivoire	4.1	6.2	9.2	21.7	44.6	85.9
Djibouti	0.3	0.4	0.6	1.3	2.6	4.7
Egypt	125.2	157.1	191.9	253.3	342.2	415.0
Equatorial Guinea	0.1	0.1	0.2	0.4	0.7	1.3
Eritrea	0.3	0.4	0.6	2.0	4.4	8.1
Ethiopia	3.8	7.7	13.0	41.3	87.9	160.1
Gabon	1.5	2.1	3.0	5.3	8.9	14.7
Gambia	0.2	0.3	0.5	1.2	2.7	5.2
Ghana	7.1	11.4	16.6	35.6	71.1	137.1
Guinea	0.6	0.6	1.1	4.1	9.6	18.0
Guinea-Bissau	0.03	0.03	0.1	0.4	1.0	1.9
Kenya	6.2	9.2	14.6	38.0	76.6	142.1
Lesotho	0.6	0.8	0.9	1.5	2.3	3.4
Liberia	0.2	0.3	0.5	1.8	4.0	7.5
Libya	25.4	24.6	30.1	39.7	50.9	65.1
Malawi	1.7	1.5	2.1	5.9	12.8	22.9
Mali	1.2	1.6	2.5	7.7	17.8	34.8
Mauritania	0.7	0.8	1.0	1.9	3.3	5.1
Morocco	23.9	32.7	39.9	52.7	67.4	86.3
Mozambique	9.8	14.1	21.6	46.8	82.3	139.8
Namibia	3.4	4.1	4.8	6.8	9.7	13.8
Niger	0.7	1.0	1.8	7.6	20.0	41.6
Nigeria	20.7	29.1	45.4	119.2	258.1	510.4
Rwanda	0.3	0.5	0.9	3.5	7.7	13.8
Senegal	2.5	3.7	5.6	13.0	27.2	53.1
Sierra Leone	0.1	0.1	0.2	1.4	3.3	5.8
Somalia	0.3	0.4	0.8	3.4	8.2	15.7
South Africa	202.8	205.6	229.9	281.6	350.5	433.7
Sudan and South Sudan	6.0	10.5	16.6	43.4	87.3	161.4
Swaziland	1.3	1.3	1.6	2.3	3.3	4.6
Tanzania	3.3	6.1	9.9	28.0	57.6	104.2
Togo	0.8	1.2	1.8	4.7	10.1	19.3
Tunisia	13.6	16.2	19.7	26.1	33.3	42.7
Uganda	1.5	2.6	4.6	16.5	37.8	72.1
Zambia	7.8	11.7	13.9	21.4	34.0	52.3

Zimbabwe	7.4	8.1	9.7	15.1	23.6	35.2
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Table 10 Electrified population 2016-2050.

Region	2016	2030	2050
	With electricity access [million]	Without electricity access [million]	With electricity access [million]
Africa	479,059,770	707,947,230	1,642,653,000
Algeria	39,912,560	463,440	48,274,000
Angola	6,180,660	19,650,340	39,351,000
Benin	3,098,890	8,067,110	15,593,000
Botswana	1,494,240	809,760	2,817,000
Burkina Faso	3,346,020	15,286,980	27,244,000
Burundi	1,194,270	10,358,730	17,357,000
Cameroon	13,399,490	10,525,510	32,947,000
Central African Rep.	130,630	4,868,370	6,490,000
Chad	524,640	13,972,360	21,946,000
Congo	1,702,980	3,038,020	6,790,000
Congo, Dem. Rep.	8,684,620	71,038,380	120,304,000
Cote d'Ivoire	6,198,380	17,055,620	32,143,000
Djibouti	453,590	446,410	1,054,000
Egypt	92,852,790	530,210	117,102,000
Equatorial Guinea	574,650	295,350	1,238,000
Eritrea	1,762,680	3,589,320	7,311,000
Ethiopia	25,403,550	76,449,450	138,297,000
Gabon	1,061,420	701,580	2,321,000
Gambia	759,140	1,295,860	3,105,000
Ghana	20,402,180	7,630,820	36,865,000
Guinea	1,607,160	11,339,840	18,276,000
Guinea-Bissau	406,540	1,481,460	2,541,000
Kenya	9,832,400	37,418,600	65,412,000
Lesotho	595,580	1,564,420	2,486,000
Liberia	69,360	4,545,640	6,414,000
Libya	6,316,550	13,450	7,418,000
Malawi	1,705,380	16,044,620	26,584,000
Mali	5,348,740	12,786,260	27,370,000
Mauritania	1,216,420	2,949,580	5,666,000
Morocco	34,406,360	410,640	39,787,000
Mozambique	11,407,710	17,343,290	41,437,000
Namibia	822,390	1,691,610	3,272,000
Niger	3,112,640	17,602,360	35,966,000
Nigeria	83,619,450	103,367,550	262,599,000
Rwanda	2,787,710	9,095,290	15,785,000
Senegal	8,624,600	6,965,400	22,802,000
Sierra Leone	331,720	6,260,280	8,598,000
Somalia	1,729,310	9,349,690	16,493,000
South Africa	21,286,810	34,232,190	64,900,000
Sudan and South Sudan	9,707,970	44,201,030	74,253,000
Swaziland	357,440	946,560	1,507,000

Tanzania	15,268,210	39,886,790	82,927,000	137,136,000
Togo	1,998,780	5,497,220	10,489,000	15,681,000
Tunisia	11,236,000	0	12,527,000	13,271,000
Uganda	6,005,490	34,317,510	61,929,000	101,873,000
Zambia	4,484,650	12,232,350	25,313,000	42,975,000
Zimbabwe	5,637,020	10,329,980	21,353,000	29,615,000

Table 11 Electricity access in Africa (2012) [5].

Region	National electrification rate %	Urban electrification rate%	Rural electrification rate %
Africa	43%	68%	26%
Algeria	99%	100%	96%
Angola	30%	46%	6%
Benin	28%	55%	6%
Botswana	66%	75%	51%
Burkina Faso	16%	54%	2%
Burundi	10%	34%	7%
Cameroon	54%	88%	17%
Central African Rep.	3%	5%	1%
Chad	4%	16%	0%
Congo	35%	52%	5%
Congo, Dem. Rep.	9%	24%	1%
Cote d'Ivoire	26%	42%	8%
Djibouti	50%	61%	14%
Egypt			
Equatorial Guinea	66%	93%	48%
Eritrea	32%	86%	17%
Ethiopia	23%	85%	10%
Gabon	60%	64%	34%
Gambia	35%	60%	2%
Ghana	72%	90%	52%
Guinea	12%	28%	3%
Guinea-Bissau	20%	37%	6%
Kenya	20%	60%	7%
Lesotho	28%	55%	17%
Liberia	2%	3%	0%
Libya	100%	100%	99%
Malawi	9%	33%	5%
Mali	27%	55%	12%
Mauritania	21%	47%	2%
Morocco	99%	100%	97%
Mozambique	39%	66%	27%
Namibia	30%	50%	17%
Niger	14%	62%	4%
Nigeria	45%	55%	35%
Rwanda	17%	67%	5%
Senegal	55%	90%	28%
Sierra Leone	5%	11%	1%
Somalia	15%	33%	4%

South Africa	85%	88%	82%
South Sudan	1%	4%	0%
Sudan	35%	63%	21%
Swaziland	27%	40%	24%
Tanzania	24%	71%	7%
Togo	27%	35%	21%
Tunisia	100%	100%	100%
Uganda	15%	55%	7%
Zambia	26%	45%	14%
Zimbabwe	40%	80%	14%

Table 12 Relative increase of the “others” electricity demand per power pool [5].

Relative increase	2012-2020	2020-2030	2030-2040
North Africa	1.34%	1.32%	1.28%
West Africa	1.75%	2.02%	1.93%
Nigeria	1.84%	2.15%	1.99%
Central Africa	1.63%	1.73%	1.64%
East Africa	1.91%	2.16%	1.86%
Tanzania	1.88%	2.00%	1.65%
Southern Africa	1.26%	1.35%	1.38%
Mozambique	1.88%	2.00%	1.65%
South Africa	1.17%	1.20%	1.22%

Appendix B: Africa energy resource potential

Table 13 Africa oil resources and reserves [4].

Country 2011	Recoverable Coal (million short tons)		Crude Oil (Billion Barrels)	Natural Gas (Trillion Cubic Feet)
	Anthracite and Bituminous (million short tons)	Lignite		
	2011	2011	2015	2015
Africa	34833.03896	235.89463	126.50311	604.144
Algeria	65.03637	0	12.2	159.054
Angola	NA	NA	9.011	9.711
Benin	NA	NA	0.008	0.04
Botswana	44.09245	0	0	0
Burkina Faso	NA	NA	0	0
Burundi	NA	NA	0	0
Cameroon	NA	NA	0.2	4.77
Central African Rep.	0	3.30693	0	0
Chad	NA	NA	1.5	0
Congo	NA	NA	1.6	3.2
Democratic. Rep. of Congo	97.0034	0	0.18	0.035
Cote d'Ivoire	NA	NA	0.1	1

Djibouti	NA	NA	0	0
Egypt	17.63698	0	4.4	77.2
Equatorial Guinea	NA	NA	1.1	1.3
Eritrea	NA	NA	0	0
Ethiopia	NA	NA	0.00043	0.88
Gabon	NA	NA	2	1
Gambia	NA	NA	0	0
Ghana	NA	NA	0.66	0.8
Guinea	NA	NA	0	0
Guinea-Bissau	NA	NA	0	0
Kenya	NA	NA	0	0
Lesotho	NA	NA	0	0
Liberia	NA	NA	0	0
Libya	NA	NA	48.363	53.183
Malawi	0	2.20462	0	0
Mali	NA	NA	0	0
Mauritania	NA	NA	0.02	1
Morocco	90.38953	44.09245	0.00068	0.051
Mozambique	233.69001	0	0	100
Namibia	NA	NA	0	2.2
Niger	77.1618	0	0.15	NA
Nigeria	23.14854	186.29062	37.07	180.49
Rwanda	NA	NA	0	2
Senegal	NA	NA	0	0
Sierra Leone	NA	NA	0	0
Somalia	NA	NA	0	0.2
South Africa	33241.30136	0	0.015	NA
Sudan and South Sudan	NA	NA	5	3
Swaziland	158.73284	0	0	0
Tanzania	220.46227	0	0	0.23
Togo	NA	NA	0	0
Tunisia	NA	NA	0.425	2.3
Uganda	NA	NA	2.5	0.5
Zambia	11.02311	0	0	0
Zimbabwe	553.3603	0	0	0

Table 14 Renewable Energy Potential by country [22] [24] [25] [26].

Country	CSP	PV	Wind CF >20%	Wind CF >30%	Geothermal
	TWh/year	TWh/year	TWh/year	TWh/year	MW
Algeria	26,530	27,904	30,155	2,535.9	-
Angola	9,786	13,319	202	-	-
Benin	-	3,898	405	-	-
Botswana	13,070	13,764	9,793	302.9	-
Burkina Faso	-	7,742	4,154	7.5	-
Burundi	786	888	-	-	n/a
Cameroon	3,706	10,105	979	15.9	-
Central African Rep.	3,471	5,284	79	-	-
Chad	10,284	10,506	9,165	1,519.4	-
Congo	2	6,778	-	-	-
DRC	12,439	22,862	2,173	41.4	n/a
Cote d'Ivoire	221	10,325	430	-	-
Djibouti	852	947	934	149.1	1,000
Egypt	26,605	32,218	36,601	6,185	n/a
Eq. Guinea	-	706	-	-	-
Eritrea	4,349	4,775	3,154	412.4	n/a
Ethiopia	22,959	27,154	14,838	3,002.1	5,000
Gabon	6	5,402	-	-	-
Gambia	316	474	173	1.3	-
Ghana	229	7,644	606	2.4	-
Guinea	467	5,204	2	-	-
Guinea-Bissau	906	1,493	124	-	-
Kenya	15,399	23,046	22,476	4,446.4	10,000
Lesotho	1,122	938	599	40.1	-
Liberia	-	667	-	-	-
Libya	11,823	13,979	21,649	5,149.5	n/a
Malawi	4,474	5,210	1,986	262.1	n/a
Mali	-	7,906	1,923	-	-
Mauritania	4,988	7,990	11,822	2,940.5	-
Morocco	15,127	15,155	11,297	1,458.8	-
Mozambique	16,851	22,024	10,805	395.9	-
Namibia	29,716	26,183	15,196	497	-
Niger	8,829	15,669	14,628	1,262	-
Nigeria	10,045	32,456	12,867	95.3	-
Rwanda	789	892	-	-	700
Senegal	1,537	7,519	5,454	323.6	-
Sierra Leone	197	1,499	-	-	-
Somalia	13,156	25,687	43,539	10,616.4	n/a
South Africa	43,275	42,243	41,195	6,076.3	-
Sudan	77,422	87,817	61,661	9,837.8	400
Swaziland	559	572	476	9.7	-
Tanzania	31,482	38,804	18,456	2,295.2	650
Togo	-	1,257	79	-	-
Tunisia	2,045	4,645	6,842	1,244	n/a
Uganda	8,582	9,470	815	100.7	450
Zambia	15,691	17,894	13,229	1,145	-
Zimbabwe	11,874	15,684	12,137	1,000.3	-

Table 15 Distance coastline factors of inland countries.

Country	Distance coastline factor
Botswana	1.6
Burkina Faso	1.7
Burundi	1.1
Central African Rep.	1.1
Chad	1.1
Ethiopia	1.5
Lesotho	1.3
Malawi	1.5
Mali	1.6
Niger	1.7
Rwanda	1.1
Swaziland	1.1
Uganda	1.1
Zambia	1.8
Zimbabwe	1.4

Appendix C: Detailed Transmission Data

Table 16 Local transmission and distribution losses.

Transmission [1]		Distribution								
		Rural [1]			Urban [1]			Others		
		2010	2020	2040	2010	2020	2040	2010	2020	2040
Algeria	5.0%	30.0%	20.0%	20.0%	20.0%	15.0%	8.0%	12.0%	9.0%	5.0%
Angola	5.0%	30.0%	20.0%	20.0%	20.0%	15.0%	8.0%	15.0%	10.0%	5.0%
Benin	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	8.0%	6.0%
Botswana	5.0%	20.0%	20.0%	20.0%	15.0%	10.0%	8.0%	9.0%	6.0%	6.0%
Burkina Faso	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	8.0%	7.0%
Burundi	1.4%	12.0%	12.0%	12.0%	1.6%	1.6%	1.6%	2.0%	2.0%	2.0%
Cameroon	1.4%	12.0%	12.0%	12.0%	1.6%	1.6%	1.6%	1.0%	1.0%	1.0%
Central African Rep.	1.4%	12.0%	12.0%	12.0%	1.6%	1.6%	1.6%	2.0%	2.0%	2.0%
Chad	1.4%	12.0%	12.0%	12.0%	1.6%	1.6%	1.6%	2.0%	2.0%	2.0%
Congo	1.4%	12.0%	12.0%	12.0%	1.6%	1.6%	1.6%	2.0%	2.0%	2.0%
Congo, Dem. Rep.	5.0%	20.0%	20.0%	20.0%	25.0%	10.0%	8.0%	15.0%	5.0%	4.0%
Cote d'Ivoire	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	8.0%	6.0%
Djibouti	2.3%	12.0%	12.0%	12.0%	3.6%	3.6%	3.6%	3.0%	3.0%	3.0%
Egypt	3.6%	12.0%	12.0%	12.0%	1.3%	1.3%	1.3%	5.0%	5.0%	5.0%
Equatorial Guinea	1.4%	12.0%	12.0%	12.0%	1.6%	1.6%	1.6%	2.0%	2.0%	2.0%
Eritrea	5.0%	30.0%	20.0%	20.0%	20.0%	15.0%	8.0%	16.0%	10.0%	8.0%
Ethiopia	3.4%	12.0%	12.0%	12.0%	4.4%	4.4%	4.4%	4.0%	4.0%	4.0%
Gabon	1.4%	12.0%	12.0%	12.0%	1.6%	1.6%	1.6%	2.0%	2.0%	2.0%
Gambia	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	15.0%	8.0%	7.0%
Ghana	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	8.0%	6.0%
Guinea	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	17.0%	4.0%	4.0%
Guinea-Bissau	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	17.0%	5.0%	5.0%
Kenya	5.3%	12.0%	12.0%	12.0%	9.6%	9.6%	9.6%	9.0%	8.0%	7.0%

Lesotho	5.0%	25.0%	20.0%	20.0%	12.0%	10.0%	8.0%	11.0%	9.0%	7.0%
Liberia	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	17.0%	3.0%	3.0%
Libya	5.0%	30.0%	20.0%	20.0%	20.0%	15.0%	8.0%	15.0%	11.0%	6.0%
Malawi	5.0%	30.0%	20.0%	20.0%	20.0%	10.0%	8.0%	14.0%	8.0%	6.0%
Mali	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	6.0%	5.0%
Mauritania	5.0%	30.0%	20.0%	20.0%	20.0%	15.0%	8.0%	17.0%	11.0%	7.0%
Morocco	5.0%	30.0%	20.0%	20.0%	20.0%	15.0%	8.0%	12.0%	9.0%	5.0%
Mozambique	5.0%	30.0%	20.0%	20.0%	30.0%	15.0%	8.0%	10.0%	8.0%	5.0%
Namibia	5.0%	25.0%	20.0%	20.0%	20.0%	10.0%	8.0%	9.0%	6.0%	5.0%
Niger	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	7.0%	6.0%
Nigeria	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	7.0%	6.0%
Rwanda	1.8%	12.0%	12.0%	12.0%	0.7%	0.7%	0.7%	1.0%	1.0%	1.0%
Senegal	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	8.0%	6.0%
Sierra Leone	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	7.0%	3.0%	3.0%
Somalia	5.0%	30.0%	20.0%	20.0%	20.0%	15.0%	8.0%	16.0%	10.0%	8.0%
South Africa		25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	7.0%	5.0%	5.0%
Sudan and South Sudan	7.0%	12.0%	12.0%	12.0%	10.5%	10.5%	10.5%	9.0%	9.0%	8.0%
Swaziland	5.0%	20.0%	20.0%	20.0%	15.0%	10.0%	8.0%	11.0%	7.0%	6.0%
Tanzania	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	14.0%	8.0%	6.0%
Togo	5.0%	25.0%	20.0%	20.0%	17.0%	10.0%	8.0%	16.0%	8.0%	6.0%
Tunisia	5.0%	30.0%	20.0%	20.0%	20.0%	15.0%	8.0%	11.0%	8.0%	5.0%
Uganda	2.3%	12.0%	12.0%	12.0%	3.5%	3.5%	3.5%	3.0%	3.0%	3.0%
Zambia	5.0%	30.0%	20.0%	20.0%	25.0%	15.0%	8.0%	12.0%	7.0%	4.0%
Zimbabwe	5.0%	20.0%	20.0%	20.0%	17.0%	12.0%	8.0%	10.0%	7.0%	5.0%

Table 17 International trade links [1].

From	To	Capacity (MW)	Investment cost (\$/kW)	Status	First year	Included in Reference Trade scenario	Comments
Algeria	Libya	1000	895	Assumed	2025	No	
Algeria	Morocco	1440	671	Assumed	2025	No	
Algeria	Mali	500	2014	Assumed	2025	No	
Algeria	Mauritania	500	2462	Assumed	2025	No	
Algeria	Niger	300	1865	Assumed	2025	No	
Angola	Botswana	1500	224	Planned	2020	No	
Angola	DRC	2100	137	Planned	2016	Partly-600MW	
Angola	Namibia	1900	240	Planned	2016	No	
Angola	Zambia	1000	1641	Assumed	2025	No	
Benin	Ghana	655	180	Assumed	2025	No	
Benin	Togo	300	142	Assumed	2025	No	
Botswana	Namibia	2100	75	Planned	2018	Partly-1500MW	
Botswana	South Africa	1300	205	Existing		Partly-800MW	
Botswana	Zambia	600	1044	Assumed	2025	No	

Botswana	Zimbabwe	1250	75	Existing		Partly-650MW
Burkina Faso	Benin	300	746	Assumed	2025	No
Burkina Faso	Mali	306	574	Planned	2015	No
Burkina Faso	Togo	300	746	Assumed	2025	No
Burundi	Rwanda	430	105	Existing		Partly-100MW
Burundi	Tanzania	100	171	Assumed	2025	No
Cameroon	Gabon	500	522	Assumed	2025	No
Cameroon	Nigeria	1000	709	Assumed	2025	No
Cameroon	Chad	125	926	Planned	2020	Yes
Central African Rep.	Congo	100	746	Assumed	2025	No
Central African Rep.	Cameroon	100	746	Assumed	2025	No
Central African Rep.	Sudan	100	2089	Assumed	2025	No
Central African Rep.	Chad	100	821	Assumed	2025	No
Cote d' Ivoire	Burkina Faso	327		Existing		Yes
Cote d' Ivoire	Ghana	982	137	Existing		Yes
Cote d' Ivoire	Guinea	500	895	Assumed	2025	No
Cote d' Ivoire	Liberia	338	177	Committed	2014	Yes
Congo	Cameroon	300	1268	Assumed	2025	No
Congo	Gabon	600	725	Planned	2020	No
DRC	Burundi	475	107	Existing		Yes
DRC	Central African Rep.	100	203	Assumed	2025	No
DRC	Congo	60		Existing		Yes
DRC	Namibia	3000	203	Planned	2020	Partly-1500MW
DRC	Rwanda	527	71	Existing		Yes

DRC	Sudan	1000	2238	Assumed	2025	No
DRC	Tanzania	1000	603	Assumed	2025	No
DRC	Uganda	500	603	Assumed	2025	No
DRC	South Africa	3000	1405	Planned	2023	Partly-2500MW
DRC	Zambia	2310	603	Existing		Partly-810MW
Djibouti	Eritrea	100	597	Assumed	2025	No
Djibouti	Somalia	100	1194	Assumed	2025	No
Egypt	Sudan	6000	517	Planned	2016	No
Equatorial Guinea	Cameroon	600	244	Planned	2020	No
Ethiopia	Djibouti	180		Existing		Yes
Ethiopia	Eritrea	200	746	Assumed	2025	No
Ethiopia	Kenya	2000	423	Planned	2018	No
Ethiopia	Sudan	6600	190	Existing		Partly-200MW
Ethiopia	Somalia	400	1044	Assumed	2025	No
Gabon	Equatorial Guinea	600	494	Planned	2020	No
Gambia	Guinea-Bissau	329	275	Planned	2022	Yes
Ghana	Burkina Faso	332	202	Planned	2017	No
Ghana	Benin	963	137	Existing		Yes
Ghana	Togo	963	137	Existing		Yes
Guinea	Mali	321	366	Committed	2020	Yes
Guinea-Bissau	Guinea	310	585	Planned	2022	Yes
Guinea-Bissau	Senegal	100	373	Assumed	2025	No
Kenya	Sudan	1000	1492	Assumed	2025	No
Kenya	Somalia	200	865	Assumed	2025	No
Lesotho	South Africa	360	50	Existing		Partly-230MW
Liberia	Guinea	338	177	Planned	2020	Yes
Liberia	Sierra Leone	303	816	Planned	2017	Yes
Libya	Egypt	740	1514	Assumed	2025	No
Libya	Niger	300	2238	Assumed	2025	No
Libya	Sudan	1000	1492	Assumed	2025	No
Libya	Chad	200	1492	Assumed	2025	No

Libya	Tunisia	1000	536	Assumed	2025	No
Malawi	Mozambique	900	117	Planned	2015	No
Malawi	Tanzania	500	895	Assumed	2025	No
Malawi	Zambia	200	416	Planned	2018	No
Mali	Cote d'Ivoire	320	428	Planned	2016	Yes
Mali	Mauritania	500	1044	Assumed	2025	No
Mali	Niger	500	1015	Assumed	2025	No
Mali	Senegal	429	288	Existing		Yes
Mauritania	Senegal	250	380	Assumed	2025	No
Morocco	Algeria	1680	597	Assumed	2025	No
Mozambique	Swaziland	1450		Existing		Yes
Mozambique	Tanzania	1000	1194	Assumed	2025	No
Mozambique	South Africa	3850	123	Existing		Yes
Mozambique	Zambia	600	1268	Assumed	2025	No
Mozambique	Zimbabwe	1200	98	Existing		Partly-700MW
Namibia	South Africa	4050	227	Existing		Partly-2550MW
Namibia	Zambia	3800	75	Existing		Partly-2900MW
Namibia	Zimbabwe	2100	220	Planned	2018	Partly-1500MW
Niger	Burkina Faso	638	393	Planned	2014	No
Niger	Benin	650	256	Planned	2014	No
Niger	Chad	200	1268	Assumed	2025	No
Niger	Togo	650	256	Planned	2014	Yes
Nigeria	Benin	1333	255	Existing		Partly-686MW
Nigeria	Niger	832	219	Existing		Yes
Nigeria	Chad	200	813	Assumed	2025	No
Nigeria	Togo	1333	255	Existing		Partly-686MW
Rwanda	Tanzania	200	118	Assumed	2025	No
Senegal	Gambia	341	106	Planned	2017	Yes
Senegal	Guinea	286	1012	Planned	2017	Yes
Sierra Leone	Guinea	334	243	Planned	2014	Yes
South Africa	Swaziland	1900	16	Existing		Partly-1450MW
South Africa	Zimbabwe	2750	220	Existing		Partly-1250MW
Sudan	Eritrea	200	671	Assumed	2025	No

Sudan	Chad	200	1768	Assumed	2025	No
Sudan	Uganda	500	1858	Assumed	2025	No
Spain	Morocco	1400		Existing		Yes
Tanzania	Kenya	1920	77	Planned	2016	Partly-1520MW
Tanzania	Uganda	759	43	Existing		Yes
Tanzania	Zambia	400	204	Planned	2017	No
Togo	Ghana	655	180	Planned	2020	Yes
Tunisia	Algeria	1680	597	Assumed	2025	No
Uganda	Kenya	858	161	Existing		Partly-418MW
Uganda	Kenya	500	161	Assumed	2025	No
Uganda	Rwanda	250		Existing		Yes
Zambia	Zimbabwe	2800	440	Existing		Partly-2200MW

Appendix D: Electricity trades among the countries

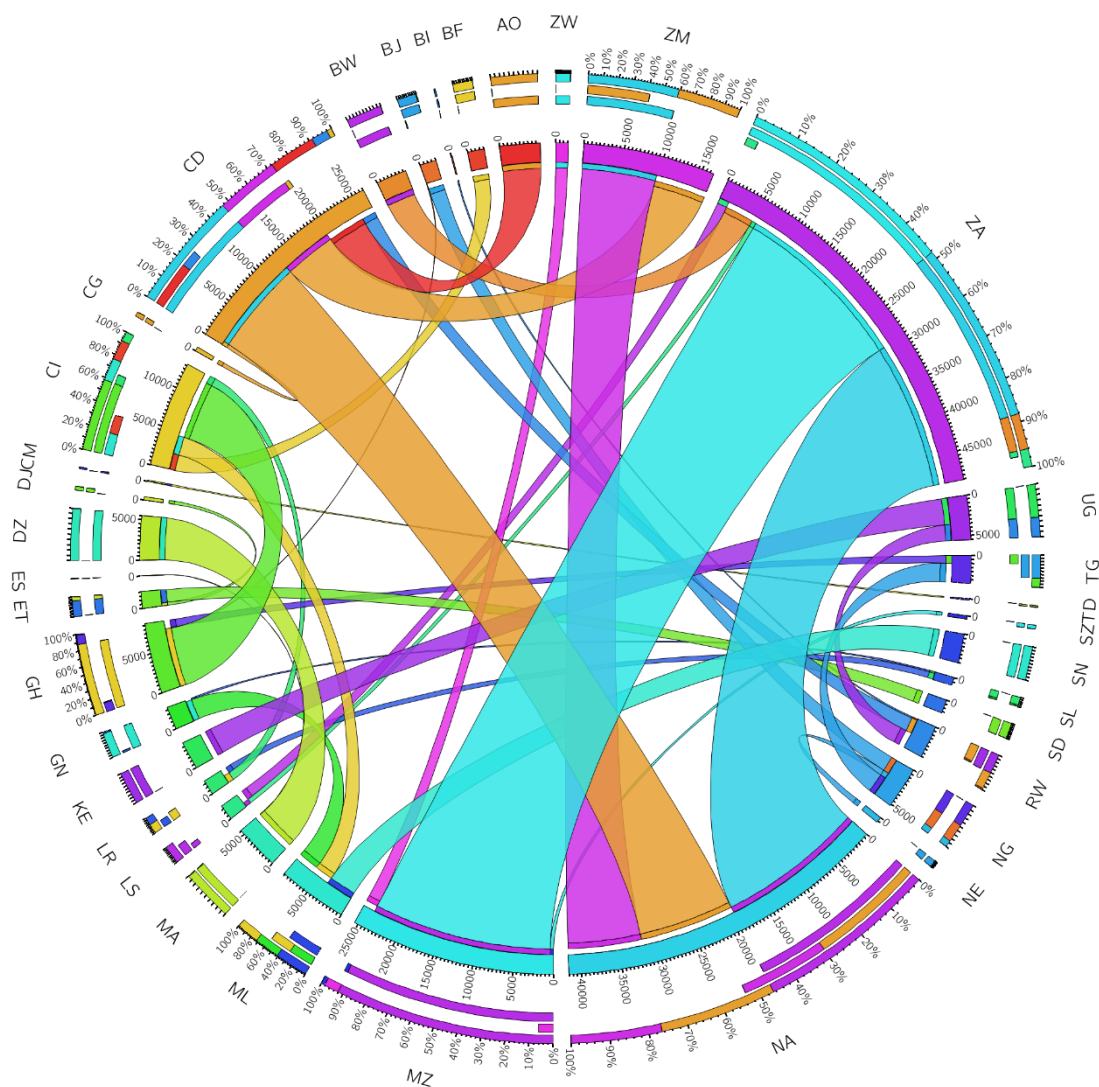


Figure 18 Electricity trades among the African countries in the Reference scenario 2020 (GWh).

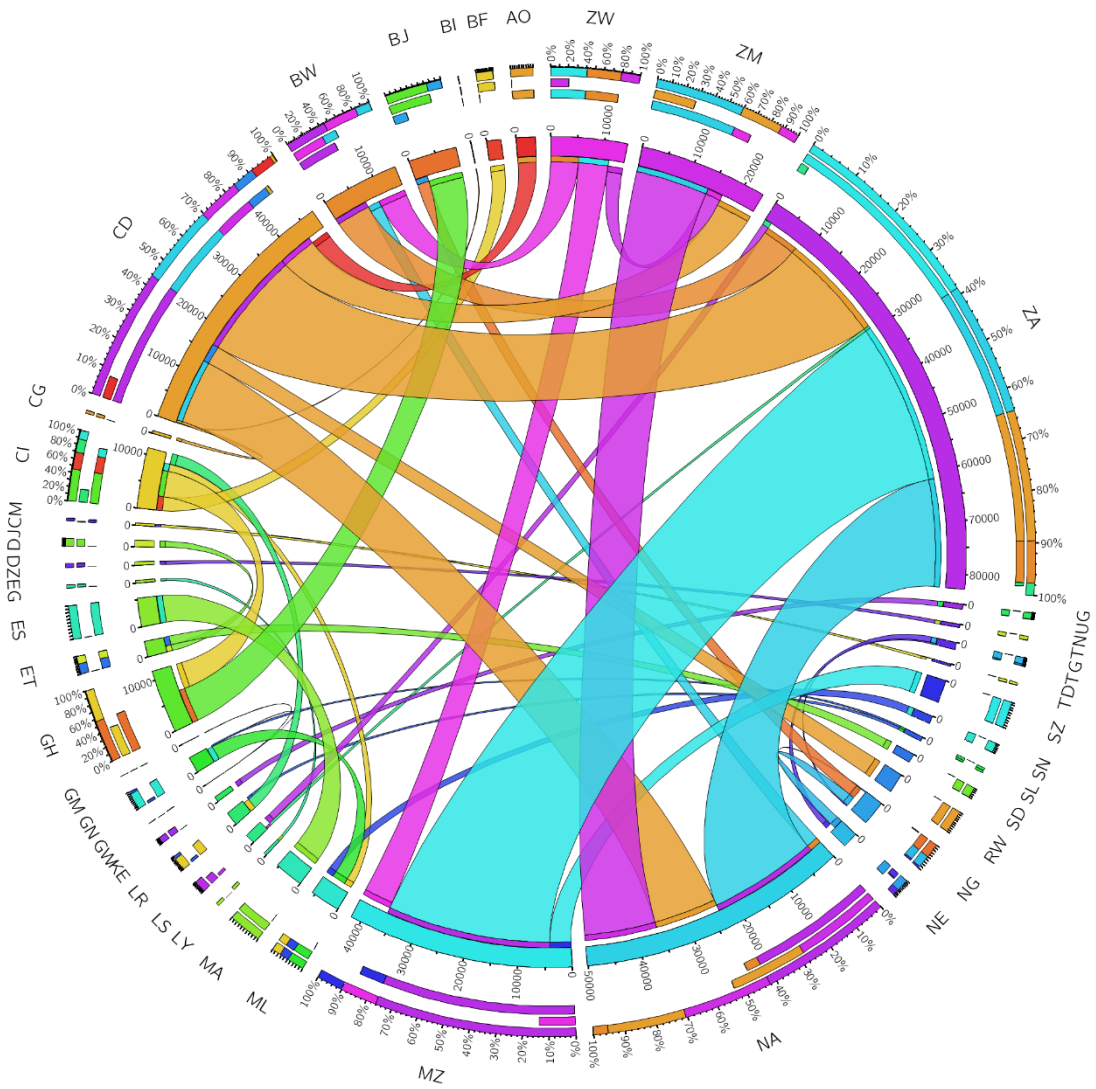


Figure 19 Electricity trades among the African countries in the Reference scenario 2030 (GWh).

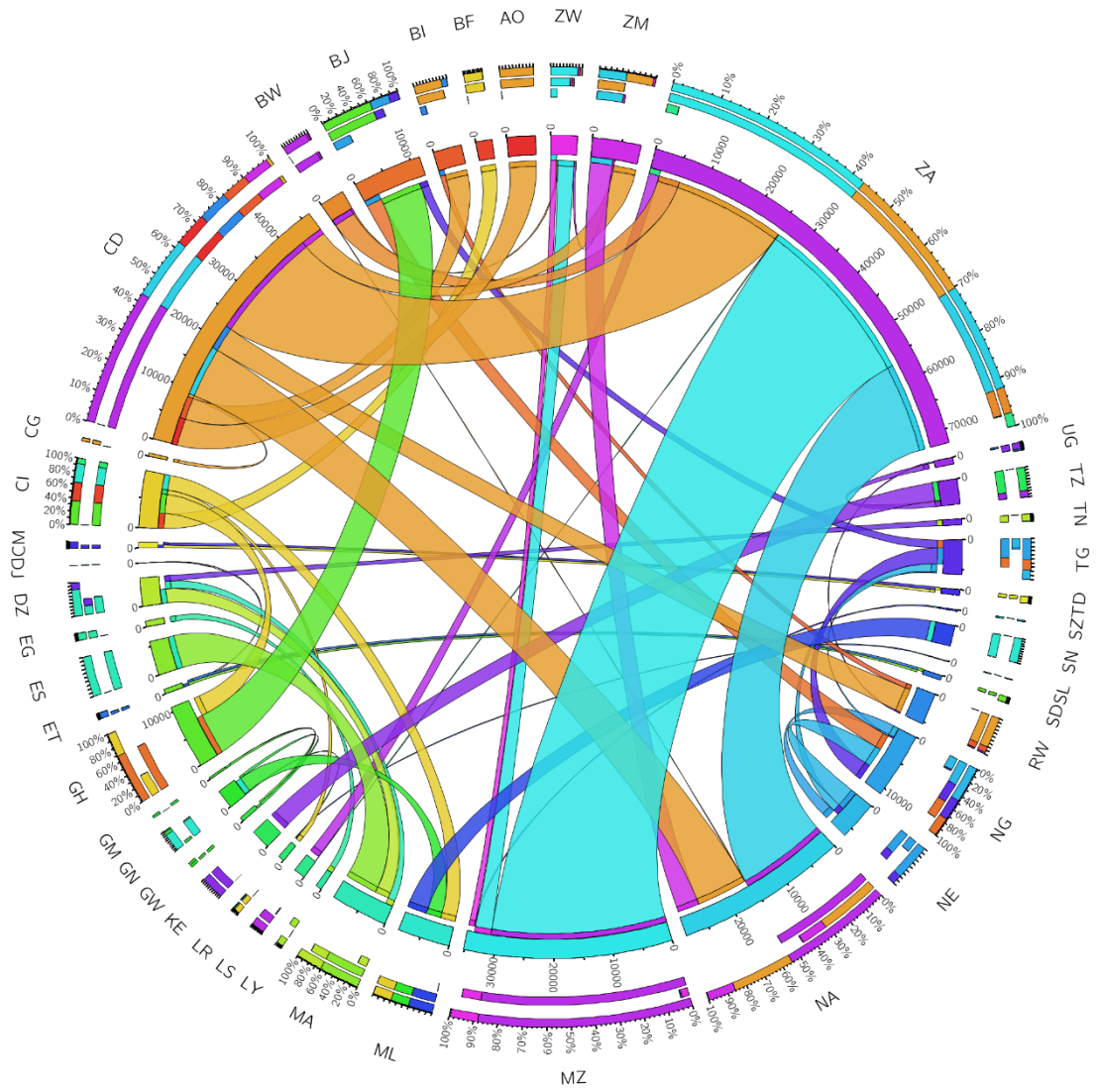


Figure 20 Electricity trades among the African countries in the Reference scenario 2040 (GWh).

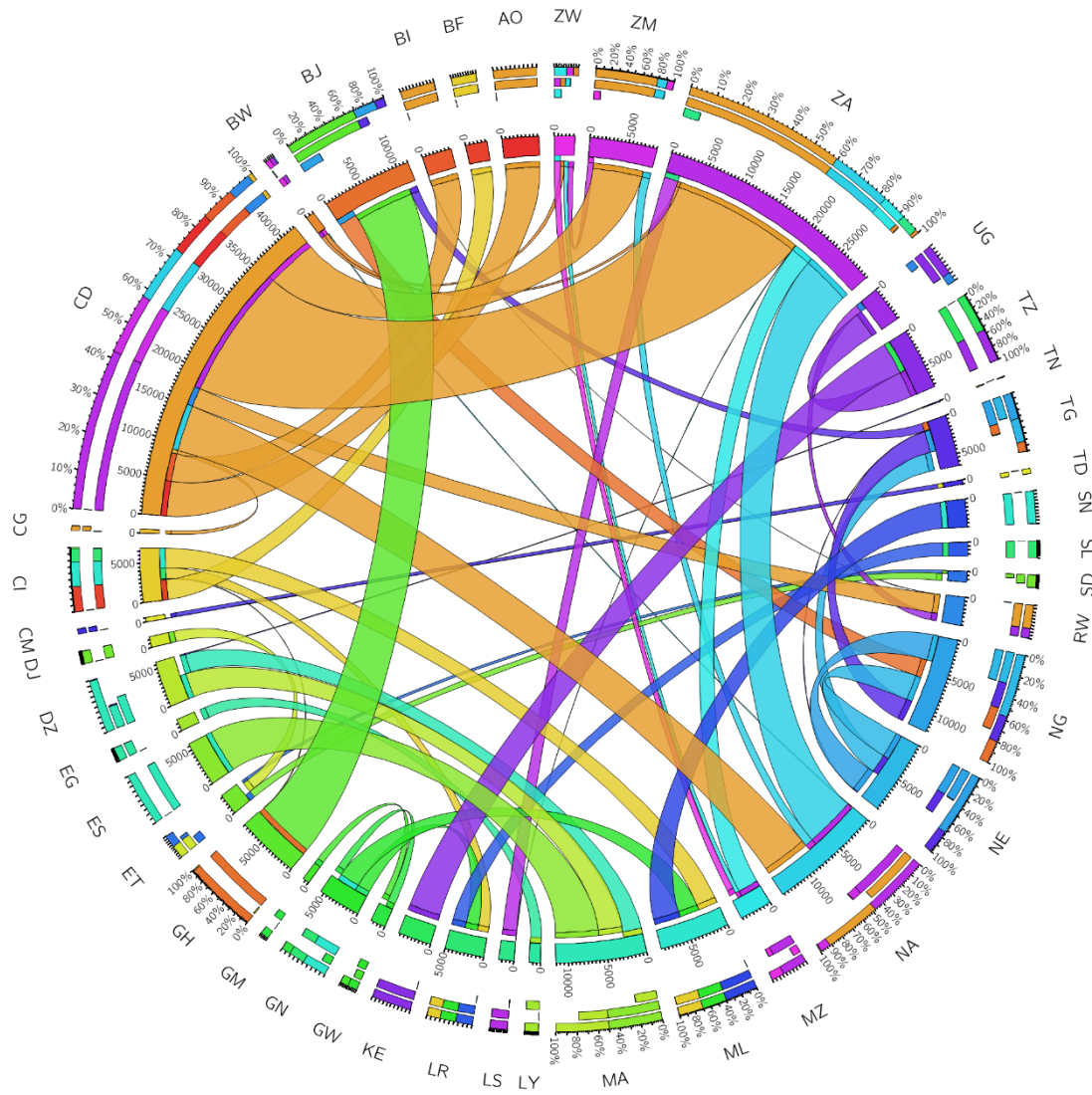


Figure 21 Electricity trades among the African countries in the Reference scenario 2050 (GWh).

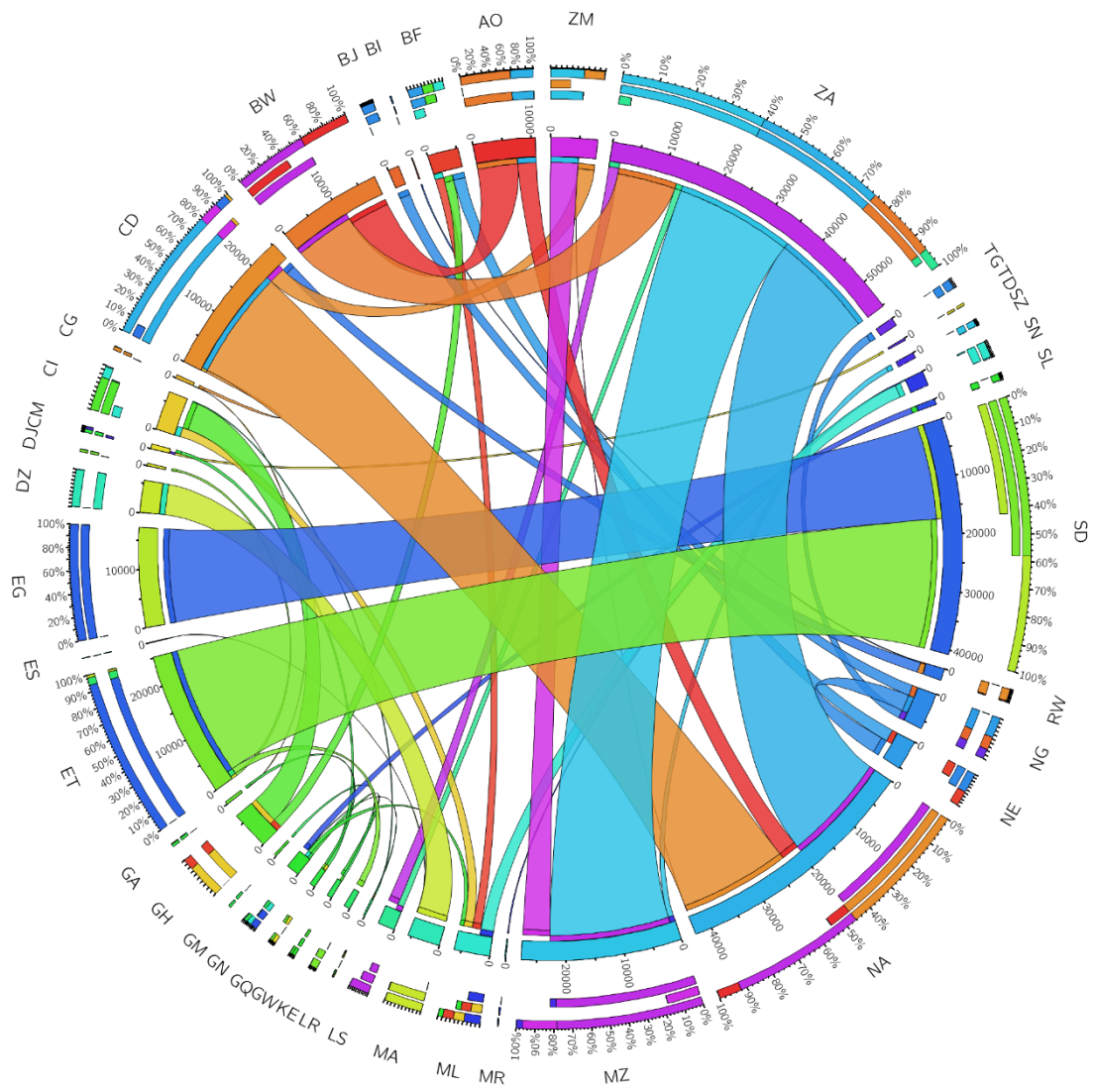


Figure 22 Electricity trades among the African countries in the Enhanced scenario 2020 (GWh).

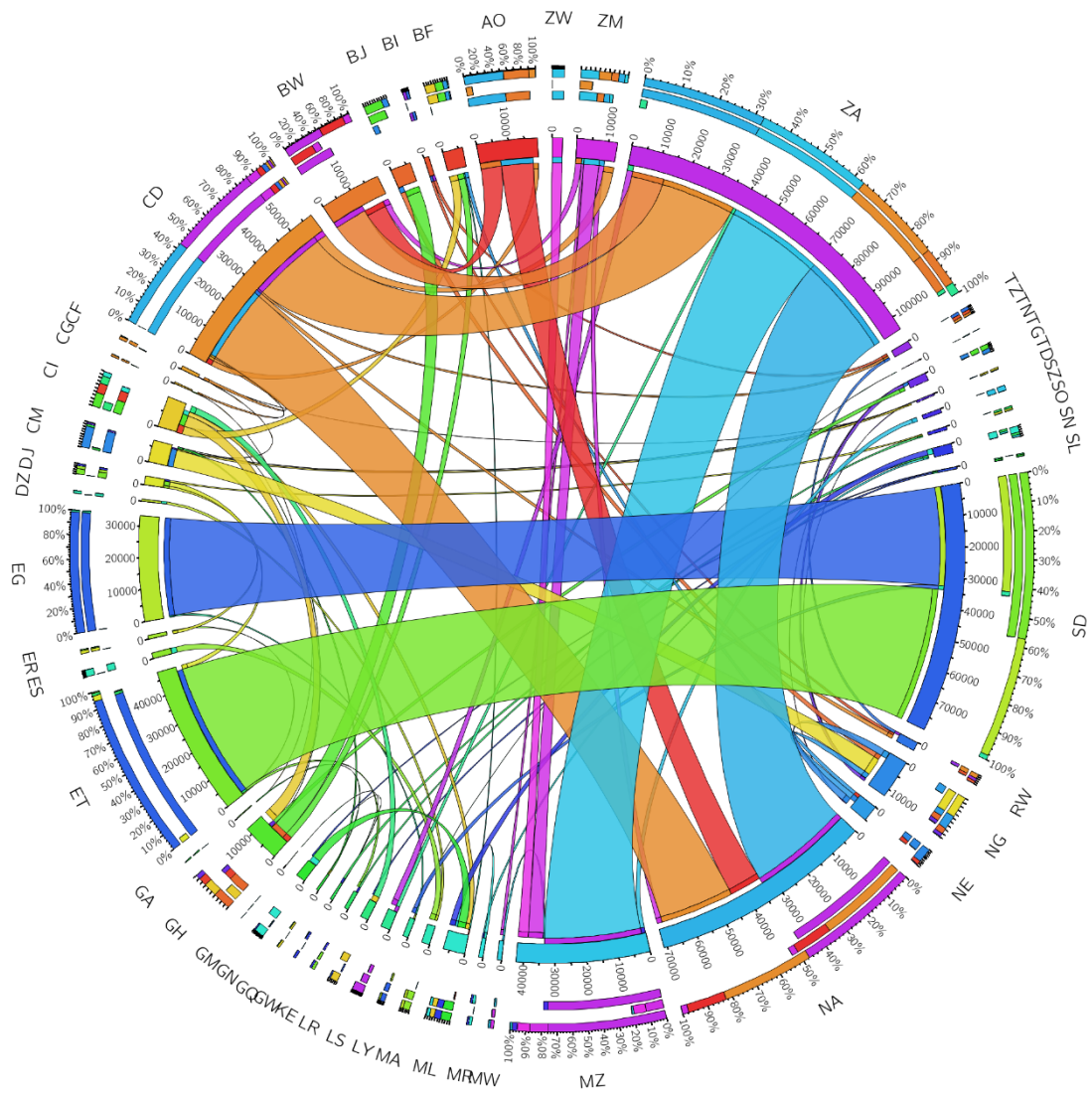


Figure 23 Electricity trades among the African countries in the Enhanced scenario 2030 (GWh).

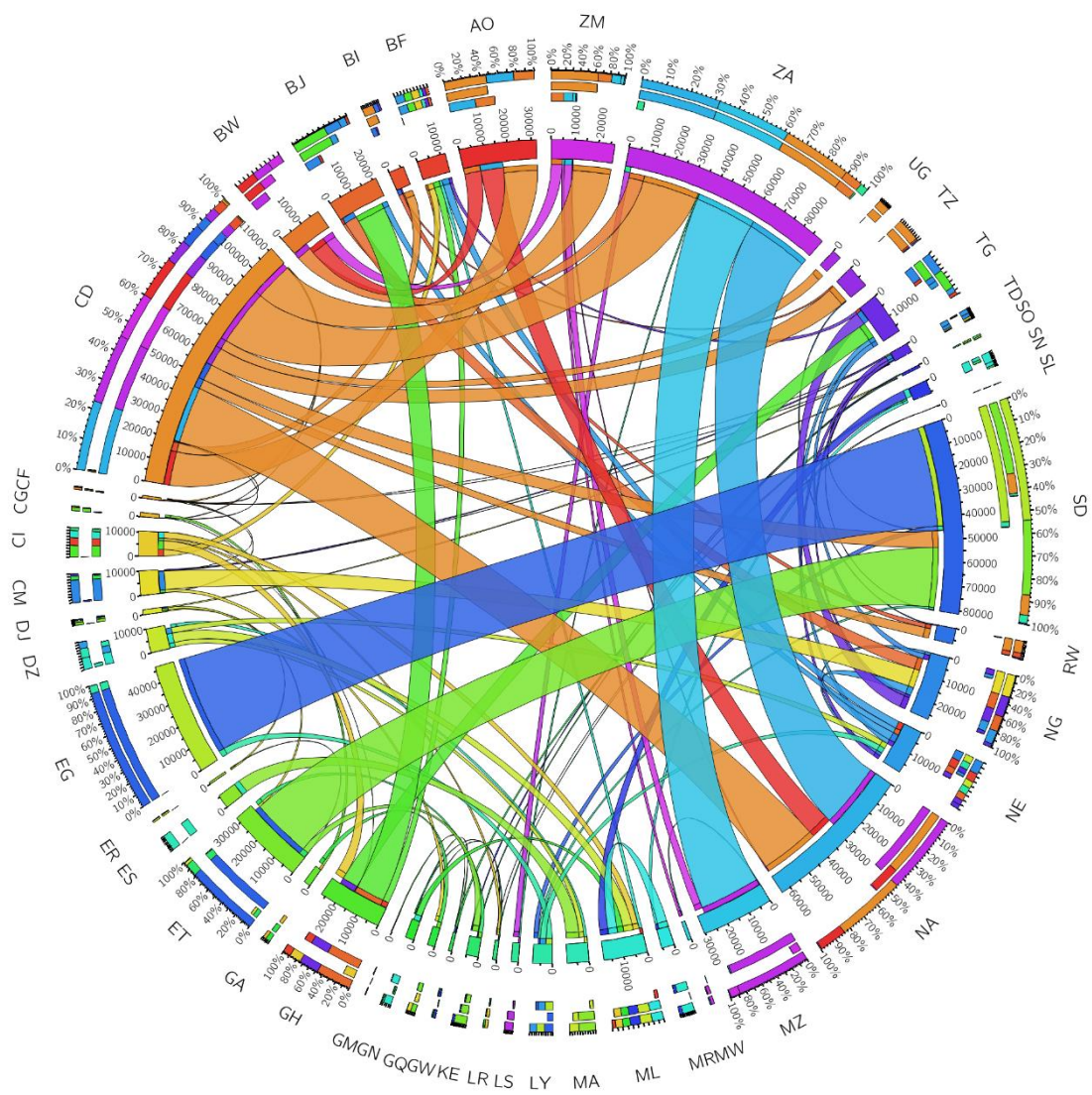


Figure 24 Electricity trades among the African countries in the Enhanced scenario 2040 (GWh).

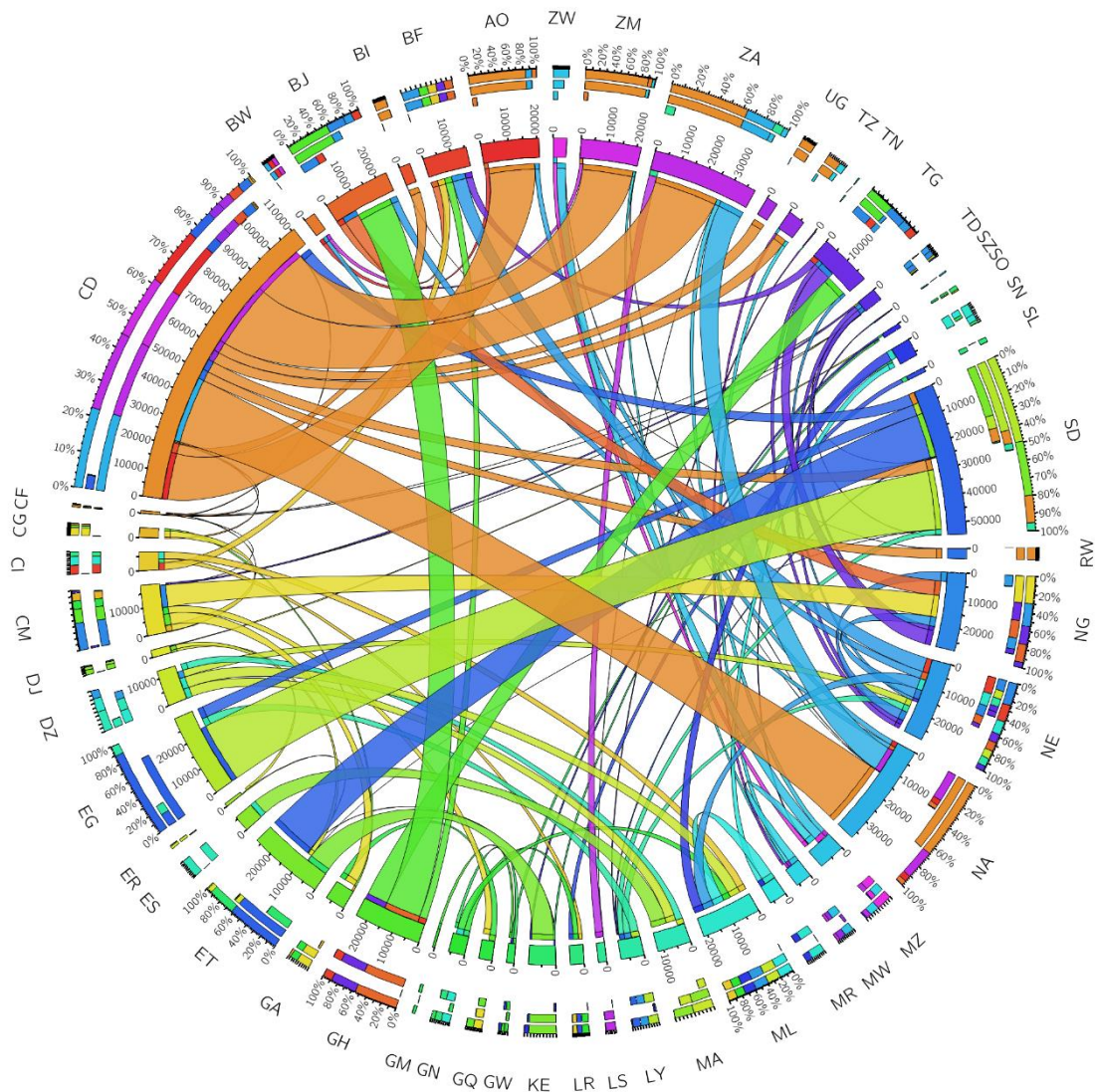


Figure 25 Electricity trades among the African countries in the Enhanced scenario 2050 (GWh).